Evidence of Structural Aliasing in Time-Series Forecasting with Chronos-Bolt-Tiny

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Executive Summary

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Time-Series Forecasting

Time-series forecasting is a fundamental time-series analysis task in which a model learns temporal dependencies and dynamic patterns linking past observations to future data in order to predict future values or trends[1].

During the preparation of this work, the authors used AI-based tools, namely GPT-5.6 Sol, Gemini 3.7 Flash, Sonnet 5 and Opus 5, in distinct roles. The tools supported language refinement and clarity throughout the report, and assisted with the implementation and refactoring of the probing, data-collection and analysis code released with this work, including the Bayesian model descriptions in Appendix D and the figure-generating scripts. The tools were not used to generate experimental results, which come from the recorded runs in the accompanying repository. All generated or suggested content was reviewed, revised where necessary and validated by the authors, who assume full responsibility for the final content of this work.

Evolution of Forecasting Methodologies

Classical time-series analysis relies on statistical models such as AutoRegressive Integrated Moving Average (ARIMA) and Exponential Smoothing (ETS). These models are simple and interpretable, but their assumptions of linearity and stationarity can limit their ability to capture nonlinear relationships and evolving data[1].

Deep learning models relax these constraints by learning nonlinear temporal representations directly from data. Recurrent neural networks process sequential dependencies through recurrent hidden states, whereas Transformers use self-attention to capture long-range temporal relationships, but attention scales quadratically with sequence length. Patch-based Transformer architectures then group neighbouring observations into tokens, reducing sequence length and giving attention access to local temporal structure[1].

Time-series foundation models sit at the end of this line. Chronos treats time-series values as discrete tokens obtained by scaling and quantisation[2]; its successor Chronos-Bolt replaces quantisation with the patching scheme above and a direct multi-quantile head[3, 4, 5]. Patching is therefore not an incidental implementation detail, but the design choice that makes this class of models practical at scale.

Classic Aliasing

Whatever the architecture downstream, predictive fidelity is bounded by the digital representation of the signal. The governing limit is the Nyquist-Shannon sampling theorem: a band-limited continuous-time signal is perfectly reconstructible only if sampled uniformly at a rate f_s strictly greater than twice its highest frequency component $f_{\max}$,

$$f_s > 2f_{\max}. \tag{1}$$

The threshold $f_N = f_s/2$ is the Nyquist frequency. When the criterion is violated by an insufficient sampling rate or a missing anti-aliasing filter, components above f_N fold back into the lower spectrum and become indistinguishable from genuine low-frequency content, an irreversible acquisition-time loss known as *aliasing*, whereas satisfying Equation 1 is conventionally treated as sufficient for a discrete sequence to represent its source faithfully.

Application Domain and Scope

The reference setting is a designed scenario: a photovoltaic array supplying an internal load, a local storage battery and the external grid, with a control agent performing load balancing, battery dispatch, state monitoring and anomaly detection on short-horizon forecasts. Measured photovoltaic and weather telemetry are taken from the SolarTech Lab dataset[6]; the concepts, states and relations of the scenario are formalised as an OWL2 ontology in Ontology Domain Description.

The target architecture is Chronos-Bolt-Tiny, with default patch size $P = 16$, stride $S = 16$, context window $T = 2048$, four encoder and four decoder blocks, and a direct quantile projection head. The objective is to test the assumption that such a model preserves the frequency content of Nyquist-compliant signals, and to characterise how f_s, P and S interact in determining when it does not.

Patch-Based Tokenization: a Formal Description

Let $\mathbf{x} = \{x[n]\}_{n\in\mathbb{N}}$, with $\mathbb{N} = \{0, 1, \ldots\}$, be a discrete-time signal representing the input time series. We define the patch-based tokenization operator $\mathcal{T}_{P,S}$ with patch size $P \in \mathbb{N}^+$ and stride $S \in \mathbb{N}^+$ as a mapping that transforms the signal $\mathbf{x}$ into a sequence of vector-valued tokens $\mathbf{T} = \{\mathbf{t}_k\}_{k\in\mathbb{N}}$, where each token $\mathbf{t}_k \in \mathbb{R}^P$.

The k-th token is defined as

$$\mathbf{t}_k = \mathcal{T}_{P,S}(\mathbf{x})[k] = \begin{bmatrix} x[kS] \\ x[kS+1] \\ \vdots \\ x[kS+P-1] \end{bmatrix}. \tag{2}$$

Using element-wise indexing notation,

$$\mathbf{t}_k[m] = x[kS+m], \qquad m \in \{0, \dots, P-1\}. \tag{3}$$

Phase Locking

Consider a continuous-time periodic signal sampled uniformly at sampling frequency f_s, yielding the discrete-time sequence $x[n]$. We isolate a fundamental harmonic component and model it as

$$x[n] = \sin\left(2\pi \frac{f}{f_s} n + \phi\right), \tag{4}$$

where f denotes the signal frequency satisfying the Nyquist criterion $f < \frac{f_s}{2}$, and $\phi \in [0, 2\pi)$ is an arbitrary phase offset.

Phase-locking occurs when the structural phase profile of the signal repeats exactly under a translational shift equal to the stride S. Evaluating a stride translation yields

$$x[n+S] = \sin\left(2\pi \frac{f}{f_s}(n+S) + \phi\right) = \sin\left(2\pi \frac{f}{f_s} n + 2\pi \frac{fS}{f_s} + \phi\right). \tag{5}$$

For exact alignment between the signal and the tokenization operator, the phase accumulation over one stride must correspond to an integer number of 2π-periods

$$2\pi \frac{fS}{f_s} = 2\pi c, \qquad c \in \mathbb{N}^+. \tag{6}$$

As a result, the signal exhibits exact stride-period translational invariance

$$x[n+S] = \sin\left(2\pi \frac{f}{f_s} n + 2\pi c + \phi\right) = \sin\left(2\pi \frac{f}{f_s} n + \phi\right) = x[n]. \tag{7}$$

Applying the same derivation to a translation of one patch size P yields

$$2\pi \frac{fP}{f_s} = 2\pi k, \qquad k \in \mathbb{N}^+. \tag{8}$$

This condition implies exact patch-period translational invariance,

$$x[n+P] = x[n]. \tag{9}$$

Solving Equation 6 and Equation 8 for f yields two distinct phase-locking frequency families,

$$\mathcal{F}_S = \left\{ c \frac{f_s}{S} \right\}_{c \in \mathbb{N}^+}, \qquad \mathcal{F}_P = \left\{ k \frac{f_s}{P} \right\}_{k \in \mathbb{N}^+}. \tag{10}$$

Each family is a comb, that is an evenly spaced set of frequencies, of spacing f_s/S and f_s/P respectively. Using the two families in Equation 10, the phase-locking frequency class $\mathcal{F}_{\text{lock}}$ is defined as

$$\mathcal{F}_{\text{lock}} = \left\{ f \in \mathbb{R}^+ \;\middle|\; f \in \mathcal{F}_S \cup \mathcal{F}_P,\ f < \frac{f_s}{2} \right\}. \tag{11}$$

It is convenient to express both conditions in units that do not depend on the sampling rate. Introducing the cycles-per-stride and cycles-per-patch ratios

$$\text{cps} = \frac{fS}{f_s}, \qquad \text{cpp} = \frac{fP}{f_s}, \tag{12}$$

Equation 6 and Equation 8 read simply cps $\in \mathbb{N}^+$ and cpp $\in \mathbb{N}^+$: $\mathcal{F}_{\text{lock}}$ is the set of frequencies completing a whole number of cycles within one stride or within one patch. This normalisation makes the two branches directly comparable across geometries and across sampling rates.

Neither branch is restricted to sinusoidal signals. Both conditions generalise to an arbitrary discrete-time signal invariant under a translation equal to the stride or to the patch,

$$x[n+S] = x[n] \qquad \text{or} \qquad x[n+P] = x[n], \qquad \forall n, \tag{13}$$

but only the first bears on consecutive patches. Tokens begin at multiples of the stride, so the step from token k to token $k+j$ is a shift of jS samples. Substituting Equation 3 and reducing by the stride condition at $j = 1$ gives

$$\mathbf{t}_{k+1}[m] = x[(k+1)S + m] = x[kS + m + S] = x[kS + m] = \mathbf{t}_k[m], \tag{14}$$

and since this holds for every component,

$$\mathbf{t}_{k+1} = \mathbf{t}_k. \tag{15}$$

Here and below, for positive integers A and B, the notation $A \mid B$ means that A divides B, namely $B = qA$ for some $q \in \mathbb{N}^+$.

The patch condition reduces the same shift only when P divides jS. At $j = 1$ that requires $P \mid S$, which under $S \leq P$ occurs only at $S = P$, so consecutive tokens are left distinct. It does hold at the smallest j for which $P \mid jS$, namely $j = P/\gcd(P, S)$; since $S/\gcd(P, S) \in \mathbb{N}^+$, repeated application of patch-periodicity one period at a time gives

$$\begin{aligned} \mathbf{t}_{k+j}[m] &= x[(k+j)S + m] \\ &= x[kS + m + jS] \\ &= x\left[kS + m + \frac{S}{\gcd(P,S)}P\right] \\ &= x[kS + m + iP], \qquad\qquad i \in \mathbb{N}^+ \\ &= x[kS + m] \\ &= \mathbf{t}_k[m], \end{aligned} \tag{16}$$

and therefore

$$\mathbf{t}_{k+j} = \mathbf{t}_k. \tag{17}$$

Patch-periodicity thus makes tokens equal as well, but at a spacing of j rather than between neighbours. The two conditions coincide on consecutive patches only in the non-overlapping geometry, where $j = 1$.

Analysis of Structural Redundancies and Invariances

Whenever $S \leq P$, every sample of the input appears in at least one token and can be read back from it: taking $k = \lfloor n/S \rfloor$ and $m = n \bmod S \leq S - 1 \leq P - 1$ in Equation 3 gives

$$x[n] = \mathbf{t}_{\lfloor n/S \rfloor}[n \bmod S], \qquad S \leq P. \tag{18}$$

Complete raw patching with $S \leq P$ is therefore injective: the token sequence determines the signal, and no frequency below Nyquist is destroyed by tokenization itself. Everything that follows concerns redundancy in the token sequence rather than loss of information, the single exception being the $S > P$ regime, where the operator does discard samples.

Case 1: Equal Patch Size and Stride ($P = S$)

When patches are contiguous and non-overlapping the two branches coincide, $\mathcal{F}_S = \mathcal{F}_P$, and $j = P/\gcd(P,P) = 1$, so Equation 17 collapses onto Equation 15. This is the only geometry in which both conditions act on consecutive patches, and every locked frequency then gives

$$\mathbf{t}_k = \mathbf{t}_0, \qquad \forall k. \tag{19}$$

Case 2: Overlapping Windows ($S < P$)

When the stride is smaller than the patch size consecutive windows share samples, and the fraction of a patch they share is the overlap ratio

$$O = \frac{P - S}{P} \in [0, 1), \tag{20}$$

which is 0 for contiguous patches and approaches 1 as the stride shortens.

In this regime the two branches separate. Since $f_s/S > f_s/P$, the stride comb is the sparser of the two. The derivation of Equation 14 makes no use of the relation between P and S, so at $f \in \mathcal{F}_S$ consecutive tokens are still exactly equal.

Case 3: Disjoint Windows ($S > P$)

When the stride exceeds the patch size the windows leave $S - P$ unobserved samples between consecutive patches, and Equation 18 fails: $\mathcal{T}_{P,S}$ is non-injective for every f, since signals differing only on the gaps share a token sequence. The loss is one of discarded samples rather than of tokenization geometry, so these configurations are excluded by construction from the experimental design.

Structural Aliasing as a Representation-Level Property

$\mathcal{F}_S$ collects the stride-lock candidates, at which consecutive raw tokens repeat. $\mathcal{F}_P$ collects the frequencies whose period divides the patch, at which the sequence repeats every $P/\gcd(P,S)$ tokens while neighbours stay distinct unless $S = P$. Their union $\mathcal{F}_{\text{lock}}$ is therefore a statement about candidate geometry, not a proof of aliasing: for $S \leq P$ the raw token sequence remains a lossless encoding of the input at every one of these frequencies, so none of the translations derived above is by itself evidence that two distinct signals become indistinguishable to the model.

Structural aliasing is the separate, representation-level proposition that a model trained on such sequences nevertheless fails to make those frequencies readable from its internal state. It is treated here as an emergent and falsifiable property of the trained representations[7], diagnosed by readability rather than by a distance between raw tokens: it is present when the frequency of the signal can no longer be recovered from the internal state at a locked frequency as well as it can at neighbouring, non-locked ones. It is accordingly a property of the learned representation rather than of the operator, and one that can only be established empirically, with $\mathcal{F}_{\text{lock}}$ supplying the candidate frequencies at which to look. The hypotheses that test it are formulated in Bayesian Analysis.

Ontology Domain Description

The ontology reproduced in Appendix A closes the loop from the signal to the decision. It represents the photovoltaic micro-grid, its telemetry and a single control agent, and it makes the tokenisation geometry a first-class object rather than a property of the model: a `pv:TimeSeriesSignal` has a `pv:PatchedRepresentation` carrying `pv:patchSize`, `pv:stride` and a derived `pv:overlapRatio`, and one signal may hold several, one per (P, S) under evaluation. Since the lock conditions are arithmetic on those same fields, the set of candidates is derivable from the knowledge base rather than hard-coded beside it, which is what lets an agent recompute it when a configuration or a sampling interval changes.

The point of contact with the Bayesian analysis is the following. An `pv:ObservabilityAssessment` does not derive a state from the geometry alone. It records the estimand, its effect size, its credible interval and its posterior probability, and assigns StructurallyAliased only to a frequency that is a lock candidate and also carries confirmed evidence of loss.

Geometry alone yields a candidate, never a verdict. Because the confirmation threshold exceeds one half, confirmed loss and confirmed preservation cannot both hold. A frequency for which preservation is positively confirmed is assigned Observable; everything else is assigned PartiallyObservable: an unfinished test, a credible interval crossing the threshold, or conflicting metrics. The assessment selects the control mode. Confirmed aliasing suspends forecast-driven battery decisions.

Bayesian Analysis

Bayesian statistics represents uncertainty about unknown quantities through probability distributions and updates prior beliefs when data are observed[8]. It is used here because the hypotheses concern the magnitude and uncertainty of structural aliasing effects, not point estimates alone: posterior distributions quantify evidence for attenuation, phase independence and parameter-tracking against predeclared decision thresholds. The candidate set $\mathcal{F}_{\text{lock}}$ of Phase Locking is arithmetic and holds for any model.

The hypotheses

Three claims are carried into the analysis, each stated so that a particular posterior refutes it.

$H1$, frequency-dependent blind spots. At a candidate frequency $f_k \in \mathcal{F}_{\text{lock}}$ the model retains less of the signal than at the two neighbouring control frequencies $f_k \pm \delta$, which are not themselves candidates. The claim has two readings that need not agree: a behavioural one, in which the forecast rebuilds the tone less faithfully, and a representational one, in which the frequency becomes harder to read out of the internal state. A loss that is absent, or not localised on the candidate set, refutes $H1$.

$H2$, independence from temporal alignment. The magnitude of that loss is a property of the alignment between the signal period and the tokenisation grid, not of the point in its cycle at which the signal is observed. A deficit varying systematically with the starting phase refutes $H2$.

$H3$, sites that follow the parameter generating them. Because $\mathcal{F}_{\text{lock}}$ is a union, the claim splits in two, one per branch, with $c, k \in \mathbb{N}^+$ the harmonic indices of Phase Locking. $H3a$: at fixed P, the sites at $f = c\, f_s/S$ move proportionally to $1/S$. $H3b$: at fixed S, the sites at $f = k\, f_s/P$ move proportionally to $1/P$. A site that stays put while the parameter supposed to generate it is varied refutes its branch.

Each claim receives its own measurement, model and estimand. All the models are ordinary regressions built the same way: a measurement, a linear predictor μ_i carrying one offset per group the observation belongs to, and a likelihood, of which one coefficient is the estimand and the rest keep it unconfounded. Table 1 gives every prior with its role, and Appendix D states each model in full. Every effect prior is centred on no effect, and its scale is varied over the sensitivity ladder $\{0.25, 0.5, 1.0\}$, sceptical, primary and wide, so that a conclusion resting on the choice of scale rather than on the data is detected as such.

$H1$

The behavioural measurement is the forecast amplitude recovery $R = A_{\text{pred}}/A_{\text{true}}$, the amplitude of the tone in the forecast divided by its amplitude in the true continuation. It is collected in matched triplets: each candidate f_k with its two controls, all three sharing the same background realisation and the same phase, which is what makes the contrast paired. Writing R_k for the recovery at the candidate and $R_\pm$ for the recoveries at $f_k \pm \delta$, the contrast is

$$d = \log(R_k + 0.01) - \tfrac{1}{2}\left[\log(R_- + 0.01) + \log(R_+ + 0.01)\right], \tag{21}$$

one number per triplet, negative when the candidate recovers less than its neighbours; the additive constant keeps the logarithm finite where a tone is not rebuilt at all. Model A fits it hierarchically,

$$d_i \sim \text{Student-}t_4(\mu_i,\ \sigma), \quad \mu_i = \beta_{c[i]} + u_{k[i]} + u_{b[i]}, \quad \beta_c \sim \mathcal{N}\Big(\bar\beta + \delta_O \widetilde{O}_c + \delta_P \widetilde{\log P}_c,\ \tau\Big), \tag{22}$$

with one offset per configuration (β_c), per candidate frequency (u_k) and per background draw (u_b), and with the tilde denoting centring on the design, so that $\bar\beta$ is the effect at the average configuration. The estimand is that population effect, read as the recovery ratio $e^{\bar\beta}$. To verify: $e^{\bar\beta} < 1$ is the attenuation $H1$ predicts, whereas a posterior at or above 1 says that candidate frequencies are not attenuated in the forecast and the behavioural half of $H1$ fails. The configuration-level slope δ_O of the same fit carries $M1$ and is read in Mitigation and Agentic Deployment.

Model B asks the same question of the representation. At each probed frequency f_c a linear probe is asked to tell a tone at $f_c - 1\,\text{Hz}$ from one at $f_c + 1\,\text{Hz}$ using a captured internal state alone, and the response L_i is the prequential codelength of that sequence of labels, in bits. A codelength is positive and right-skewed with a variance growing in its mean, so the regression is Gamma with a log link,

Model	**Parameter**	**Prior**	**Role**
A, C	$\bar\beta$	Student-$t_4(0, 0.5)$	population phase-lock effect, on the log-recovery scale
A	δ_O	Student-$t_4(0, 0.5)$	slope in $\widetilde{O}_c$, the centred overlap; one unit $= 0.5$ of O
A	δ_P	Student-$t_4(0, 0.5)$	slope in $\widetilde{\log P}_c$, the centred log patch size
A, C	$\tau, \sigma_k, \sigma_b, \sigma$	Half-Student-$t_4(0, 0.5)$	configuration, harmonic, background and residual scales
C	σ_ϕ	Half-$\mathcal{N}(0, 0.25)$	spread of the eight per-phase offsets
B	α_0	$\mathcal{N}(\log \bar y, 1)$	baseline log codelength
B	θ_{lock}	$\mathcal{N}(0, 0.5)$	log codelength expansion at a lock
B	r	Gamma$(2, 0.1)$	Gamma shape (dispersion)
B	$\sigma_{st}, \sigma_{geo}$	Half-$\mathcal{N}(0, 1)$	spread of the probe-stage and geometry offsets
D1	$\alpha_g, \theta_S, \theta_P$	$\mathcal{N}(0, 1)$	per-geometry off-grid level and the two grid labels
D1	σ	Half-$\mathcal{N}(0, 1)$	residual scale of $\log z_g(f)$
D2	κ_S, κ_P	$\mathcal{N}(0, 1)$	measured spacing per unit of predicted spacing
D2	σ_F	Half-$\mathcal{N}(0, 5)$ Hz	scatter beyond the sweep-resolution floor Δf

Table 1: *Every parameter with its prior and its role. Every effect prior is centred on no effect, so a posterior that has not moved away from it is itself a report of an absent effect; the patch-size slope δ_P is estimable only because the design of Table 2 varies P at fixed S.*

$$L_i \sim \text{Gamma}(r,\ r/\mu_i)\,, \qquad \log \mu_i = \alpha_0 + \theta_{\text{lock}}\,\text{IsLocked}_i + s_{\text{st}[i]} + g_{\text{geo}[i]}, \tag{23}$$

in the shape–rate parameterisation, so that r carries the dispersion and $\mathbb{E}[L_i] = \mu_i$. Here IsLocked_i is the indicator that f_c belongs to $\mathcal{F}_{\text{lock}}$, and s_{st} and g_{geo} are zero-mean offsets for probe stage and geometry, which absorb the level differences between stages. The log link makes the estimand multiplicative: $e^{\theta_{\text{lock}}}$ is the factor by which the codelength of a candidate frequency expands. To verify: $e^{\theta_{\text{lock}}} > 1$ is the representational cost $H1$ predicts, while θ_{lock} near zero says the frequency remains as readable at a candidate as anywhere else.

$H2$

$H2$ uses the same triplets with one change. Model A pools them across phases without a phase term, so the phase is marginalised and cannot be inspected; $H2$ requires it to be kept, so the phase index is retained and the response becomes the per-phase deficit $y_i = \log(R_{\text{ctrl},i} + 0.01) - \log(R_{\text{lock},i} + 0.01)$, which is Equation 21 with its sign reversed and is positive when the candidate is worse. Model C is Model A on that response, without the configuration-level covariates, which are not what $H2$ asks about, and with the phase circle cut into eight equal slices, each given its own offset u_p:

$$y_i \sim \text{Student-}t_4(\mu_i,\ \sigma), \qquad \mu_i = \beta_{c[i]} + u_{k[i]} + u_{b[i]} + u_{p[i]}, \qquad u_p \sim \mathcal{N}(0, \sigma_\phi). \tag{24}$$

The estimand σ_ϕ is the spread of those eight offsets, which reduces the hypothesis to a single number and presupposes no shape for a possible dependence. To verify: σ_ϕ near zero means the offsets are indistinguishable and the deficit is set by the geometry alone, as $H2$ predicts, whereas a large σ_ϕ means it depends on where in its cycle the signal starts. Dropping u_p gives a nested null, which is compared with the full model by leave-one-out cross-validation (LOO); the comparison separates the size of any phase effect from the evidence that one exists.

$H3$

Both branches are tested on the token collapse $z_g(f)$, the dispersion, in geometry g and at frequency f, of the input-patch embeddings across the patches of one context; it is zero when consecutive patches are identical. All geometries are swept on one common grid, the union of a uniform one-hertz sweep with the predicted sites of every configuration in the design, so that no geometry is scored only where its own theory predicts a dip.

The first question is where the dips are located. Each swept frequency carries two binary labels, one per branch, recording whether it lies on the stride grid $\{c\, f_s/S\}$ or on the patch grid $\{k\, f_s/P\}$:

$$\log z_g(f) \sim \mathcal{N}(\alpha_g + \theta_S \cdot \text{OnStrideGrid} + \theta_P \cdot \text{OnPatchGrid},\ \sigma)\,, \tag{25}$$

with α_g the per-geometry off-grid level, and the label coefficients θ_S and θ_P as estimands. To verify: $H3$ predicts both negative, and the two-label fit should win the LOO comparison against each label alone and against neither; coefficients near zero mean the dips bear no relation to either grid.

The second question is whether the dips move, since Equation 25 remains compatible with dips that merely sit on a grid fixed once and for all. Every detected dip is assigned to the branch that predicts it and sites predicted by both are set aside, since the union of two combs has two interlaced spacings rather than one. The fundamental $\hat{f}^F_{1,g}$ of branch F in geometry g, its lowest detected site, is then compared with the spacing that branch predicts, $\Delta^S_g = f_s/S_g$ or $\Delta^P_g = f_s/P_g$:

$$\hat{f}^F_{1,g} \sim \mathcal{N}\left(\kappa_F\, \Delta^F_g,\ \sqrt{\sigma_F^2 + \Delta f^2}\right). \tag{26}$$

The slope κ_F is the estimand, measured spacing per unit of predicted spacing, and the residual scatter σ_F is given the floor Δf, the sweep step, because a spacing read off a discrete grid is no more precise than one

Run	**Patch** (P)	**Stride** (S)	**Overlap** (O)	**Tokens**	$S \mid P$	$P \mid S$
1	8	8	0.00	256	yes	yes
2	16	8	0.50	255	yes	**no**
3	16	12	0.25	170	**no**	**no**
4	16	16	0.00	128	yes	yes
5	24	8	0.67	254	yes	**no**
6	24	12	0.50	169	yes	**no**
7	24	16	0.33	127	**no**	**no**
8	24	20	0.17	102	**no**	**no**
9	24	24	0.00	85	yes	yes
10	32	8	0.75	253	yes	**no**
11	32	12	0.62	169	**no**	**no**
12	32	16	0.50	127	yes	**no**
13	32	20	0.38	101	**no**	**no**
14	32	24	0.25	85	**no**	**no**
15	32	32	0.00	64	yes	yes

Table 2: *The fifteen configurations carried into the analysis, with the overlap ratio O of Equation 20 and the number of patch tokens at the 2048-sample training context. All are trained from scratch under the same 100k-step budget and from the same seed (42); the published `chronos-bolt-tiny` checkpoint is not used. The last two columns are the divisibility conditions that decide what a geometry can attribute: $S \mid P$ means every stride site cf_s/S is also a patch site kf_s/P, so the geometry yields no stride-only frequency, and $P \mid S$ is the converse. Both hold exactly when $P = S$, where the two branches are the same set of frequencies and nothing can be attributed to either alone.*

step. To verify: $\kappa_F = 1$ denotes a branch that tracks its own arithmetic, and any other value refutes that branch. Identification here is a property of the design and not of the model, since κ_S requires a series holding P fixed while S varies and κ_P the converse; Table 2 supplies both. The geometries, the signals and the order in which the analysis executes are set out in Signal Generation, Experimental Setup and Workflow.

The mitigation claim

Signal Generation, Experimental Setup and Workflow

Testing the hypotheses of Bayesian Analysis requires a design in which the two branches of $\mathcal{F}_{\text{lock}}$ can be told apart, signals whose frequency content is known by construction, and an order of execution in which nothing that decides a hypothesis is chosen after the data have been seen. This section fixes all three.

The grid of geometries

Since any change to P or S requires retraining, the design is a grid of variants trained from scratch rather than a set of published checkpoints, each described by its patch size, its stride and the overlap ratio O of Equation 20. Configurations with $S > P$ are excluded by construction: they leave $S - P$ unobserved samples between consecutive patches, so their loss is one of discarded samples and not of tokenisation geometry, as Phase Locking sets out.

Table 2 lists the retrained runs, and two properties of the grid do the identifying work. First, P and S vary independently: series at fixed P with varying S ($P \in \{16, 24, 32\}$) isolate the stride, while series at fixed S with varying P ($S \in \{8, 12, 16\}$) isolate the patch. This is what lets Models A, D1 and D2 separate the two branches and estimate the overlap and patch-size covariates without confounding them. Second, the divisibility columns decide whether a geometry can attribute anything at all: where $S \mid P$ every stride site is also a patch site and no stride-only frequency exists, and at $P = S$ the two combs coincide entirely. Six of the fifteen runs have $S \nmid P$, and those are the ones on which θ_S and κ_S rest. All share the same training budget and seed, so a difference between them is a difference of geometry and not of training.

Signal generators

Four kinds of signal appear in this study. Three are synthetic and are what the models are probed with, each serving a different purpose rather than being a variant of the others; the fourth is the real series the findings are meant to reach.

Single-harmonic sinusoid, $x(t) = A \sin(2\pi f t + \phi)$, sampled at f_s. It is the primitive the design rests on: frequency and phase are exact, so a locked frequency produces exactly the degeneracy derived in Phase Locking and no background content can be blamed for a loss observed at a predicted site.

Light TSMixup, a controlled probing variant of the mixup augmentation used to train Chronos[2], in which the real source datasets of the original are replaced by a pool of sinusoids. It tests whether the effect survives when the tone is embedded in a multi-component signal whose composition is still fully known. Its pseudocode and parameters are given in Appendix B.

KernelSynth, the Gaussian-process generator of the same training recipe. Its output carries trends and broadband content as well as periodicity, so it tests whether the effect survives in signals resembling the corpus the model was trained on rather than the probe used to measure it. Its pseudocode, kernel bank and parameters are given in Appendix C.

Neither generated family is merely an input: each background realisation enters Models A and C as a grouping level, so a disagreement between the two would be absorbed by the background scale σ_b and made visible rather than averaged away.

Photovoltaic telemetry, the measured power and weather series of the SolarTech Lab dataset[6] that define the application domain of Ontology Domain Description. It is the target of the analysis and not one of its inputs: the hypotheses are tested on the three synthetic families above, because only there is the frequency content known by construction and a loss attributable to a named frequency. What connects the two is the normalisation of Phase Locking. The experiments run at $f_s = 512\,\text{Hz}$ whereas the telemetry is sampled once per minute, $f_s = 1/60\,\text{Hz}$, so the two share no frequency in hertz; but a candidate is the same object in both, a component completing a whole number of cycles within one stride or within one patch. A result stated at $\text{cpp} = k$ therefore transfers unchanged, and on the one-minute series with $P = 16$ it names a component of period $16/k$ minutes, the timescale at which the consequences are read in Mitigation and Agentic Deployment and Security Risk Assessment.

Experimental protocol

All synthetic signals are sampled at $f_s = 512\,\text{Hz}$ and analysed over the band $[2, 250]\,\text{Hz}$, strictly inside the Nyquist limit $f_s/2 = 256\,\text{Hz}$, so every tone probed is fully representable under the Nyquist–Shannon theorem and a loss observed at a predicted site cannot be attributed to undersampling. This is what separates the effect under test from classical aliasing. A single seed controls all stochastic generation.

The complete experimental grid is built before any data are collected: one row per combination of geometry, candidate frequency, background type and phase. Where a tone rides on a background it does so at a fixed signal-to-noise ratio of 4, the tone amplitude against the unit standard deviation of the background, so the tone whose fate is tracked is always identifiable. Each candidate frequency f_k is paired with two control frequencies $f_k \pm \delta$, the offset δ being the largest the design allows without either control landing on another member of $\mathcal{F}_{\text{lock}}$; a site for which no such offset exists is dropped rather than compared against a contaminated control. Phases are drawn from the non-redundant set available at each frequency, since a tone at f admits only finitely many distinct alignments with the sample grid.

The Bayesian fits share a fixed 480-sample probing context, which has to close on a whole number of strides: otherwise the final patch is internally padded and the padding rather than the geometry would produce a collapse. A geometry whose stride does not divide it is swept descriptively but not fitted.

Decisions and validation

Every threshold in Table 3 is fixed before a posterior is inspected, and validation comes before interpretation. Beside each rule the table records the probability that the rule already scored under the prior, so that a posterior counts as evidence only in so far as it has moved away from that value. A claim is supported at 0.95, refuted at 0.05, and otherwise reported as inconclusive, which is a verdict in its own right.

Interpretation is gated by three preregistered convergence diagnostics, and competing formulations are compared by leave-one-out cross-validation under a reading rule that returns an inconclusive verdict for differences within twice their standard error. The diagnostics, the reading rule and the sampler settings shared by every fit are stated in Appendix D; a fit failing any diagnostic is not used to support a claim.

Experimental workflow

The analysis executes in a fixed order, and the order is itself part of the design: every quantity that decides a hypothesis is specified before the model that could be tuned to it is loaded. Each stage writes its output to disk before the next begins, so that an interrupted run resumes without any statistical choice being revisited.

Exploratory frequency sweep

The first stage is descriptive and uses no Bayesian machinery. Pure sinusoids are swept across the $[2, 250]$ Hz band of the protocol above in one-hertz steps, and two measurements are taken of every tone in every geometry: the dominant frequency of a recursive multi-step forecast, which records what the model outputs, and the cross-validated accuracy of a linear probe on the output quantile head, which records what its internal state retains. The two may disagree, and a disagreement is diagnostic, since a frequency that is no longer reconstructed but remains readable implicates the output head rather than the tokenisation.

This stage decides nothing on its own, since it averages over phase, uses a single signal family and reports no uncertainty. Its purpose is to make the case for the measurement that follows: to establish, before the paired design is committed to, that some frequencies are harder to represent than others, and to reproduce on the retrained geometries the reading through which the phenomenon was first reported[7]. Both readings are given in Appendix E and summarised in Bayesian Results.

Design and priors

Priors are fixed before any observation exists. The skeleton of the experiment, one planned row per geometry, candidate frequency, background realisation, generator and phase, is constructed from the design alone, and against it are declared the priors of Table 1, the thresholds of Table 3 and the sensitivity ladder $\{0.25, 0.5, 1.0\}$ of Bayesian Analysis. Simulating responses from those priors on that skeleton, with no model

Claim	Rule	Prior	Reads as
$H1$ supported	$\Pr(\bar{\beta} < \log 0.8 \mid D) \geq 0.95$	0.34	at least 20% attenuation at a lock
$H1$ refuted	$\Pr(\lvert\bar{\beta}\rvert < \log 1.1 \mid D) \geq 0.95$	0.14	any effect is smaller than 10%
$H2$ supported	$\Pr(\sigma_\phi < \log 1.1 \mid D) \geq 0.95$	0.30	phase moves the deficit by under 10%
$H3$ location	two-label fit wins LOO, $\theta_S, \theta_P < 0$	—	dips sit on both predicted grids
$H3a$, $H3b$	$\Pr(\lvert\kappa_F - 1\rvert < 0.1 \mid D) \geq 0.95$	0.05	that branch tracks its own parameter
$M1$ supported	$\Pr(\delta_O < 0 \mid D) \geq 0.95$, overlap term wins LOO	0.50	overlap reduces the deficit

Table 3: *Decision rules.* ***Prior*** *is the probability each rule already scored under the priors of Table 1: a posterior counts as evidence only in as much as it has moved away from that value. A claim is supported at* 0.95*, refuted at* 0.05*, and otherwise inconclusive.*

output involved, constitutes a prior predictive check on the design actually used: it verifies that the priors are centred on no effect, that they admit effects large enough not to be shrunk away, and that the decision thresholds are reachable from either side.

Observation

This is the only stage that loads a trained network, and each geometry is loaded, measured and released in turn. Four families of measurement are collected, each answering a distinct question.

- **Forecast amplitude recovery.** For every triplet the median forecast is compared with the true continuation of the same signal, and the ratio of the fitted amplitudes at the frequency of interest is recorded together with the phase error. Because the candidate and its two controls share a background realisation and a phase, their recoveries can be differenced into Equation 21, the behavioural evidence about $H1$. The phase index is retained rather than averaged away, which is what makes $H2$ testable.
- **Frequency-local codelength.** The representational measurement is a minimum description length probe in prequential form[9]. **What is probed** is the internal state at ten points of the network: the register token after each of the four encoder blocks, the decoder state after each of the four decoder blocks, the pre-projection vector and the output quantile head. **What is observed** is one codelength per cell, a cell being one probe stage, one geometry and one probed frequency f_c, on the binary task of separating a tone at $f_c - 1\,\mathrm{Hz}$ from one at $f_c + 1\,\mathrm{Hz}$. The examples of a cell are split into successive chunks: the first is charged at the uniform rate of one bit per example, and each later chunk is encoded by a probe trained only on the chunks before it, at a cost of $-\log_2$ of the probability assigned to the true label. The total is the cost of transmitting the labels together with the probe that predicts them, so a probe that memorises noise pays for itself and gains nothing, which is why description length is preferred to accuracy. The task has constant difficulty at every frequency and the folds hold out phases rather than frequencies, so cells are comparable across the band. A frequency whose neighbourhood has become linearly indistinguishable inside the network costs more bits whether or not the forecast recovers it: this is the representational evidence about $H1$.
- **Across-patch token dispersion.** Before any transformer block, the input patches of one context are embedded and the spread of those embeddings across patches is recorded. It is zero exactly when consecutive patches embed to the same vector, the degeneracy the stride condition predicts, and on a generated background it appears as a deep dip instead. It is swept on the common grid formed by the union of a uniform one-hertz sweep with the predicted sites of every geometry, so that each geometry is scored where a rival theory predicts a dip as well as where its own does. This is the location evidence for $H3$.
- **Detected sites and their spacing.** The dips of each profile are located, adjacent detections merged, and each site assigned to the branch that predicts it, with sites predicted by both set aside. The fundamental of each branch, its lowest detected site, and the median spacing between successive sites are the movement evidence for $H3a$ and $H3b$.

The generated backgrounds are archived alongside the measurements, so that the exact inputs behind any posterior can be reloaded; a Gaussian-process draw is reproducible only given its seed.

Likelihood validation

The likelihood is validated before it is trusted. Contrasts are simulated on the real design structure, with the population effect fixed at four known values spanning no attenuation, the 20% reporting threshold, 50% and 90%. Each scenario is refitted and its posterior compared with the value that generated it. This distinguishes the two readings that a posterior centred on zero would otherwise admit: an effect that is absent, and a design unable to detect one.

Posterior inference

Every model is fitted under identical sampler settings, so that none receives a more or less careful exploration than another. For each hypothesis the credible interval of the estimand, the posterior probability of the preregistered threshold and the probability of practical equivalence are reported. Where competing formulations are defined, they are compared by leave-one-out cross-validation: Model A across covariate sets, which is where $M1$ is decided; Model B with and without its offsets; Model C with and without the per-phase offsets; and Model D1 across the four grid labellings, stride only, patch only, both and neither.

Checks and verdicts

Convergence is verified for every fit against the diagnostics of Appendix D. Posterior predictive checks are run on Model A, pooled and then stratified by configuration and by generator, because a hierarchical model can reproduce the marginal distribution of the contrast while systematically missing a stratum. Prior sensitivity is tested on the same model by refitting it across the sensitivity ladder and with a Gaussian likelihood in place of the Student-t_4; the conclusion is reportable only if the decision probability and the credible interval remain stable across all four variants, since otherwise what is reported is the prior rather than the data. Only then is the three-way rule of Table 3 applied to each hypothesis, and the resulting verdict is set beside the outcome that an independent frequentist test reaches on the same phenomenon.

Bayesian Results

The models of Bayesian Analysis are fitted on two sets of runs: the *clean* set, which is the fifteen configurations of Table 2, and the *full* set of twenty-two, which adds seven further geometries with shorter or odd strides that are excluded from that table. Posteriors are from the clean set unless stated otherwise; qualitative conclusions hold on both.

The exploratory sweep

Before any model was fitted, the five geometries of the initial design, (P, S) equal to $(8, 8)$, $(16, 8)$, $(16, 12)$, $(16, 16)$ and $(24, 24)$, were swept with pure sinusoids at one-hertz steps, as Signal Generation, Experimental Setup and Workflow describes; both readings are reproduced in Appendix E. Neither is a test: both average over phase and neither reports uncertainty.

Reconstruction is poor everywhere. Only 12–27% of the swept tones return a dominant frequency within 2 Hz of the input, and the root-mean-square error between output and input frequency is 127–133 Hz in every geometry, so wide stretches of the band collapse onto a few output frequencies whatever the tokenisation. The representation discriminates more finely. Pooled over the five geometries, the local $f \pm 1\,\text{Hz}$ probe scores 0.77 within one hertz of a predicted site against 0.85 elsewhere, and the share of frequencies below the 0.6 threshold doubles, from 10.4% off the candidate set to 21.0% on it. The gap holds in four of the five geometries and is widest in the two non-overlapping ones.

This is a marginal comparison, confounded with frequency and with the reduced training budget, which is why what follows pairs each candidate with matched controls and blocks on probe stage and geometry instead of reading the sweep directly. What the sweep establishes is only the premise: something in the band is harder to represent, and it is not uniformly distributed.

Validation

A parameter-recovery test on Model A fixes $\bar{\beta}$ at four known values $(0, \log 0.8, \log 0.5, \log 0.1)$. Every 95% credible interval covers the truth, and the posterior standard deviation shrinks from the prior width of 0.71 to 0.04–0.18; the full set passes with tighter intervals (0.04–0.11).

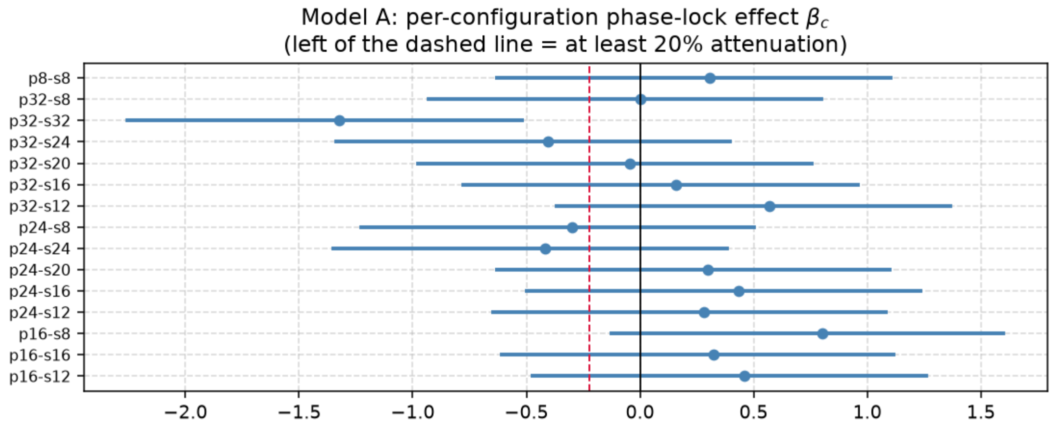

Figure 1: *Forest plot of the per-configuration phase-lock effect β_c (Model A, clean set). Each bar is the 95% credible interval; the dashed line at $\log 0.8$ marks 20% attenuation. The wide, bidirectional intervals explain the inconclusive population estimate.*

Models D1 and D2 pass every convergence criterion ($\hat{R} \leq 1.003$, ESS > 1000, zero divergences). Models A and C do not ($\hat{R} > 1.3$, ESS < 12), and Model B falls short on ESS (676 bulk clean, 525 full). No posterior from a non-converged model is used to *support* a claim at the 0.95 threshold.

H1: Behavioural and representational effects

Behavioural (Model A)

The population effect is $e^{\bar{\beta}} = 1.08$ [0.45, 2.49] (clean) and 1.02 [0.61, 1.70] (full). The probability of at least 20% attenuation is 0.19 and 0.21; the ROPE probability is 0.13 and 0.25: the verdict is **inconclusive**. Figure 1 shows the per-configuration posteriors spanning both sides of zero with no systematic direction.

Representational (Model B)

The codelength expansion at a locked frequency is $e^{\theta_{\text{lock}}} = 1.27$ [1.23, 1.31] (clean, $\Pr > \log 1.2 = 1.00$, **supported**) and 1.22 [1.19, 1.24] (full, $\Pr = 0.84$, **inconclusive**). Both agree on direction and approximate size; the difference is whether the lower tail clears the 20% mark.

The split

The behavioural test finds no consistent forecast attenuation; the representational test finds a consistent internal cost. This split — *representation penalised, forecast intact* — is the central finding: the model encodes a locked frequency less efficiently yet still reconstructs it, presumably because the decoder draws on enough redundant context to compensate.

H2: Phase invariance

The posterior is $\sigma_\phi = 0.033$ [0.020, 0.067] (clean) and 0.031 [0.019, 0.067] (full), well inside the ROPE at $\log 1.1 \approx 0.095$. The ROPE probability is 0.997 in both runs, against a prior of 0.30: the verdict is **supported**. Model C has convergence issues ($\hat{R} > 1.6$), but the margin is wide enough (0.997 vs 0.95) that the conclusion survives. Phase randomisation is therefore not a viable mitigation.

H3: Token-collapse sites

All D1 models converge ($\hat{R} \leq 1.003$, ESS > 5000). The stride-only label (M_S) wins LOO for TSMixup; the two-label model (M_{SP}) wins for pure sinusoids. In every comparison $\theta_S < 0$, confirming that frequencies on the stride grid have lower token dispersion. Figure 3 shows the dips sitting on the stride grid and shifting when the stride changes.

The stride scaling slope is $\kappa_S = 1.000$ [0.992, 1.008] (clean) and 1.006 [0.990, 1.022] (full), with $\Pr(|\kappa_S - 1| < 0.1) = 1.00$ in both runs against a prior of 0.048: $H3a$ is **supported**. The patch branch ($H3b$) is **not identified**: only 9 out of 180+ rows carry a valid fundamental, because in most configurations $S \mid P$ and no observation can be attributed to the patch branch alone.

Sensitivity and concordance

Model A is refitted across the prior-scale ladder $\{0.25, 0.5, 1.0\}$ and with a Gaussian likelihood (Figure 4). The clean-set median of $\bar{\beta}$ moves from -0.05 to $+0.29$ to -0.31; the full set shows the same instability ($+0.06$ to -0.19 to $+0.17$). The 20%-attenuation probability ranges from 0.00 to 0.62 (clean) and 0.05 to 0.43 (full). The H1 behavioural conclusion is prior-dependent and cannot be reported as data-driven.

Where Bayesian and frequentist analyses address the same question, they agree: $H1$ behavioural inconclusive, $H2$ supported, $H3a$ supported. The well-converged models (D1, D2) deliver the firmest findings; the poorly converged ones (A, B, C) deliver qualified ones.

Mitigation and Agentic Deployment

The proposed mitigation

One mitigation is carried through the design and pre-registered as $M1$: increase patch overlap by reducing the stride S at fixed patch size P and thus raising the overlap ratio O of Equation 20. Its geometric basis lies in the stride branch, whose candidates satisfy $f = cf_s/S$. In the analysed band, $c < S/2$, so their number is $\lceil S/2 \rceil - 1$. A smaller S increases the spacing and lowers this count; halving S doubles the spacing, retaining only the even-indexed sites. $M1$ asks whether this reduction also decreases the observed recovery deficit. Model A′ of Bayesian Analysis tests the claim through the overlap slope δ_O; the pre-registered rule requires $\Pr(\delta_O < 0 \mid D) \geq 0.95$.

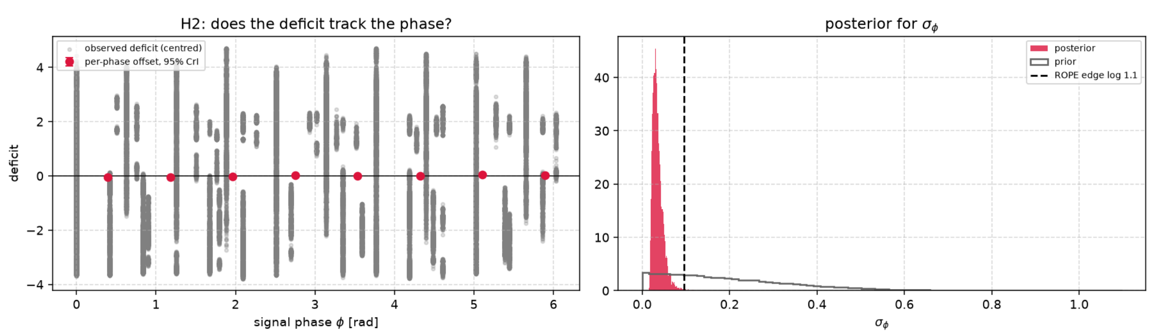

Figure 2: *Posterior for the phase term of $H2$. Left: observed deficit against phase ϕ with per-phase offsets u_p. Right: posterior for σ_ϕ against its prior, the dashed line marking the ROPE edge. Model C has not met convergence thresholds, so the posterior is provisional.*

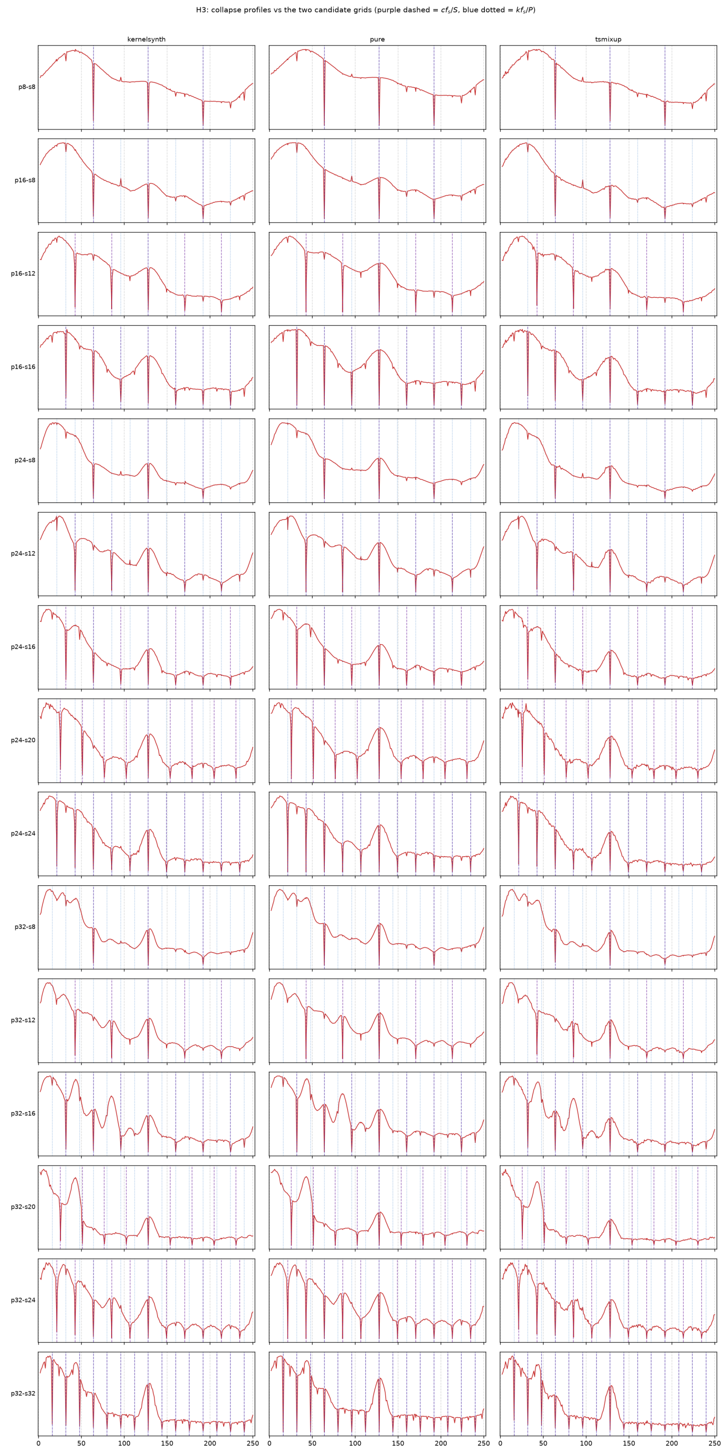

Figure 3: *Collapse profiles against the two candidate grids. Rows are configurations, columns generators; the stride grid cf_s/S is dashed and the patch grid kf_s/P dotted. Dips that follow the dashed grid as the stride changes belong to the stride branch.*

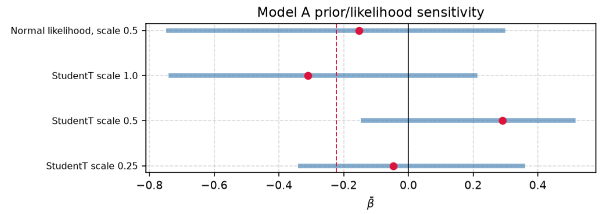

Figure 4: *Prior and likelihood sensitivity for Model A. The posterior of $\bar{\beta}$ is shown for three prior scales and for the Gaussian variant. The median shifts sign across variants.*

Comments on the Bayesian results

The posterior of δ_O is 0.44 $[-0.03, 0.81]$ on the clean set and 0.38 $[0.01, 0.72]$ on the full one. In both fits, most posterior mass lies above zero, opposite to the direction specified by the pre-registered decision rule. The corresponding probabilities, $\Pr(\delta_O < 0 \mid D) = 0.036$ and 0.023, are far below the 0.95 threshold; under that rule, $M1$ is therefore **refuted** rather than inconclusive.

Two interpretations are consistent with that result. For a context of length L, the number of complete tokens is $N = 1 + \lfloor (L - P)/S \rfloor$. Overlap and sequence length therefore change together, and their effects are not separately identified in this design. Independently, overlap thins only the stride branch while leaving the patch branch unchanged. As S decreases, the fitted slope may reflect a changing mixture of stride- and patch-derived candidates rather than an unambiguous causal effect of overlap.

The geometry also limits overlap even if the Bayesian test had favoured $M1$. Reducing S raises the stride fundamental f_s/S, whereas the patch fundamental f_s/P remains fixed with P. In the dimensionless terms of Phase Locking, overlap shifts the cycles-per-stride comb, but leaves the cycles-per-patch comb invariant. For the one-minute photovoltaic telemetry, where $f_s = 1/60\,\mathrm{Hz}$ and $P = 16$, the patch-branch candidates have periods of $16/k$ minutes and remain for every stride; overlap is therefore only a partial geometric adjustment: it displaces stride-derived candidates but cannot remove the patch-derived exposure.

In a photovoltaic-array cyber-physical system, the residual concern is configuration-dependent loss of observability at those periodicities. They must remain documented and monitored upstream of tokenisation; overlap alone cannot be credited as a complete mitigation of that exposure.

Consequences for the control agent

That conclusion determines what the agent of Appendix A can and cannot do. The ontology makes the tokenisation geometry a first-class object rather than a property hidden inside the model, so the candidate set is derivable from (f_s, P, S) by arithmetic on the knowledge base, and an `pv:ObservabilityAssessment` can mark a frequency as a candidate the moment a configuration is declared. Detection is therefore cheap and requires no experiment.

Acting on it is the harder half. The agent forecasts with Chronos, and the only mitigation available inside that model has just been refuted, so it cannot repair the forecast it depends on. What remains is honest abstention: the agent raises an alert and degrades its own authority rather than continuing to act on a signal it cannot vouch for. A confirmed StructurallyAliased assessment forces the safe mode and suspends forecast-driven battery dispatch; a merely PartiallyObservable one reduces the battery cap and lowers decision confidence. The value the agent adds here is not correction but the refusal to propagate an unwarranted forecast into an actuator.

CHATGPT ACADEMIC PROPOSAL The finding delineates the operational competence of the control agent specified in Appendix A. By treating tokenisation geometry as an ontological entity rather than an implicit model attribute, the knowledge base permits the candidate set to be derived directly from (f_s, P, S). Consequently, an `pv:ObservabilityAssessment` may identify a frequency as a candidate when the configuration is instantiated. This screening step is computationally inexpensive and does not require empirical testing.

The principal limitation concerns intervention. Because the agent relies on Chronos forecasts and the analysis establishes no in-model correction that removes the patch branch, it cannot restore information that the forecaster may fail to represent. Its appropriate response is therefore calibrated abstention: it issues an alert and reduces its decision authority instead of basing actuation on inadequately supported forecasts. A confirmed StructurallyAliased assessment selects safe mode and suspends forecast-driven battery dispatch, whereas PartiallyObservable selects a reduced battery cap and lower decision confidence. The agent's contribution is thus not the correction of the forecast, but the containment of epistemic uncertainty before it propagates to an actuator.

From representation to operational response

Taken together, the formal and Bayesian analyses establish an evidence-to-action workflow in which representation, monitoring, explanation and intervention operate as successive stages rather than independent capabilities. Patch size, stride and the derived overlap ratio are first recorded as properties of each `pv:PatchedRepresentation`. Since a signal may be evaluated under several geometries, this associates the potential structural-aliasing exposure, namely frequency-specific loss of observability, with the relevant signal–configuration pair rather than with Chronos in the abstract. The declared geometry then provides the basis for monitoring: the stride-derived sites can be recomputed from f_s/S whenever the sampling interval or stride changes, as confirmed by $H3a$, while the patch-derived sites follow analytically from f_s/P; only the former scaling was empirically identified here. A checkpoint change requires recomputation when it changes the associated (P, S) geometry. Monitoring must be applied to the raw telemetry before tokenisation, because a detector placed downstream would inherit the blind spot it is intended to identify.

A geometric candidate does not by itself establish a loss. The Bayesian layer therefore converts candidates into evidence-bearing assessments: each `pv:ObservabilityAssessment` records the estimand, effect size, credible interval and posterior probability, allowing an operator to inspect why a frequency receives a particular status. The three-way decision rule is retained in the semantics, with PartiallyObservable denoting an inconclusive or incomplete assessment rather than forcing uncertainty into either confirmation or rejection. That status then determines mode selection and carries the remaining uncertainty into actuation through a bounded, reversible adjustment of control authority. The four stages thus form a single chain: configuration identifies where degradation may occur, pre-tokenisation monitoring establishes whether relevant content is present, Bayesian evidence qualifies the observability claim, and the control policy limits the consequences when that evidence is insufficient.

A direction for future work

Since the deficit cannot be removed inside a single patch geometry, the natural next candidate is not to repair the forecaster but to route around it. The nulls of a patch tokeniser are determined by its own (f_s, P, S), so two models with different geometries are blind at different frequencies, and a third with no patch grid at all, a classical spectral or state-space forecaster, is blind at neither. An ensemble in which the agent selects the forecaster per frequency, using the observability assessment it already computes as the switch, would leave no frequency served only by a model known to be blind there. The semantic layer needed to arbitrate such a choice is the one described above, so the addition is a second forecaster rather than a second architecture for the agent. Its cost is that the two forecasters must be calibrated against each other, and the claim remains untested here: establishing it would require the same paired contrast of Bayesian Analysis applied to the ensemble output rather than to a single model.

Security Risk Assessment

The forecaster is not a standalone artefact. It sits inside a control loop in which its output drives battery dispatch and curtailment on a photovoltaic micro-grid, so a forecasting failure can become a control failure, and the property tested in this report has to be read as a security property and not only as an accuracy one. The NIST AI Risk Management Framework[10] is the reference adopted for that reading, because it separates the trustworthiness characteristics evidenced here, validity and reliability, security and resilience, from the governance functions that turn evidence into a control, and because it asks that known limitations be mapped, measured and managed rather than merely disclosed.

The limitation is a weakness before it is a vulnerability

Characterising the limitation correctly determines everything that follows. Phase-locking is deterministic, attacker-independent and present at deployment time. It is not introduced by an adversary and it is not a defect in the training data; it is a consequence of the tokenisation geometry, and it is closed-form. The candidate set follows from (f_s, P, S) alone, and those three numbers are configuration, not weights. Structural aliasing is the phenomenon that may then appear in the learned representation, and phase-locking is necessary for it but not sufficient, so the candidate set bounds where the model can be blind while Bayesian Results is what establishes where it is.

An adversary therefore needs no model access, no gradients, no query budget and no knowledge of the training corpus to enumerate that set, and only observation of the deployed forecaster to find which of its members carry a deficit. In the taxonomy of adversarial machine learning[11] this places the weakness with the availability violations rather than with integrity or privacy, and it places the attack outside the white-box assumption, which is what a conventional threat model tends to miss because it reasons about perturbations of the input tensor rather than of the physical process.

A weakness becomes a vulnerability only when someone can reach it. For the leading threat model, adversarial masking, the five elements are the following. **Adversary capabilities**: no model access of any kind and no knowledge of the training corpus, only the ability to drive a periodic load on the plant, which an inverter switching schedule or a switchable load already provides. **Attack surface**: the telemetry stream upstream of tokenisation, reached by acting on the physical process rather than on the data path. **Manipulation of telemetry**: a small periodic component at a candidate frequency, which is legitimate plant behaviour and survives any plausibility check on the raw signal. **Operational consequence**: the component is present in the plant and under-represented in the model, so dispatch and curtailment are decided on an incomplete picture, and a detector reading the forecast residual is looking through the same tokeniser that lost it. **Residual risk**: overlap thins the stride branch alone, so the patch branch remains at every deployed configuration.

What the evidence supports and what it does not

The results of Bayesian Results sharpen this picture in one direction and weaken it in another, and reporting both is the point of having measured rather than asserted.

Three findings strengthen it. The representational cost at a candidate frequency is confirmed, at a 27% codelength expansion with a credible interval well clear of the threshold, so the precondition of every scenario below is established rather than assumed. The sites track their own arithmetic exactly, $\kappa_S = 1.000\,[0.992, 1.008]$, so an adversary or an auditor can compute the target set in closed form from configuration alone and be right about where it sits. And the deficit does not depend on the phase of the signal, which removes any need for the adversary to control alignment and simultaneously removes phase randomisation from the operator's list of defences.

One finding weakens it, and it is the most consequential. The behavioural half of $H1$ came out inconclusive and, worse, prior-dependent: the forecast is not demonstrably attenuated at a candidate frequency. The

ID	Risk	AI RMF	Evidence in this report	Control available	Residual
$T1$	Adversarial masking: a periodic component injected on a candidate frequency is under-represented and escapes forecast-residual detection	MAP MEASURE	Precondition confirmed ($e^{\theta_{lock}} = 1.27$, sites track f_s/S with $\kappa_S = 1.000$); the behavioural pathway is **not** established (inconclusive and prior-dependent)	Spectral monitor on the raw stream, upstream of tokenisation, at the enumerated candidate set	high
$T2$	Systematic blind spot with no adversary: a legitimate plant harmonic sits on a candidate frequency and is persistently under-represented	MAP MEASURE MANAGE	Confirmed at the representational level; candidate set is closed-form in (f_s, P, S)	Choose (P, S) against the known spectral content of the plant; PartiallyObservable degradation	medium
$T3$	Configuration drift: a change of sampling interval, patch size or checkpoint silently relocates the blind set	GOVERN MAP	$H3a$ supported: the sites move exactly as f_s/S predicts, so relocation is deterministic and predictable	Recompute the candidate set from the knowledge base at every configuration change	low
$T4$	False assurance: overlap is increased and treated as a fix	GOVERN MANAGE	$M1$ refuted: $\delta_O = 0.44\,[-0.03, 0.81]$, the wrong sign; the patch branch is invariant to S by construction	Do not credit overlap as a control; record the refutation in the model card	high
$T5$	Defence placed inside the blind spot: phase randomisation, or anomaly detection on forecast residuals	MANAGE	$H2$ supported ($\Pr = 0.997$): the deficit does not depend on phase, so randomising it changes nothing	Move detection upstream of tokenisation; treat residual-based detection as insufficient alone	medium

Table 4: *Risk register.* ***AI RMF*** *names the function of the framework[10] under which each item is handled.* ***Residual*** *is the exposure that survives the control in the adjacent column, judged qualitatively:* ***high*** *where no control available to an operator closes the exposure,* ***medium*** *where a control exists but is partial or requires design-time choices,* ***low*** *where the exposure is closed by a procedure that follows from the arithmetic. Every entry is anchored to a posterior reported in Bayesian Results rather than asserted, and the two entries whose residual is* ***high*** *are so because the evidence is negative: $T1$ because the behavioural pathway is unproven, $T4$ because the pre-registered mitigation failed.*

strongest form of the threat model, in which the injected component is simply absent from the forecast, is therefore **not** supported by this evidence. What is supported is the weaker and still security-relevant statement that the internal representation carries a measurable, predictable and configuration-derived cost at an enumerable set of frequencies, while the decoder appears able to compensate for it under the conditions tested. A representation that is degraded but compensated is a brittleness rather than a demonstrated exploit: it is a margin that a distribution shift, a longer horizon or a lower-amplitude component could consume, and none of those were tested.

$T1$ in Table 4 is therefore carried as a plausible but unproven pathway with a confirmed precondition, not as a demonstrated attack. What would settle it is specific: the same paired contrast at the amplitude, horizon and sampling interval of the deployed telemetry rather than of the probe, on a model that meets the convergence thresholds.

Threat models beyond the adversarial one

The framework asks for systematic vulnerability, not only for attacks, and the more likely exposure has no adversary in it. On one-minute telemetry, the arithmetic of Phase Locking puts the patch branch of a $P=16$ tokeniser at periods of $16/k$ minutes, that is at 16, 8, 5.3, 4, 3.2, 2.7 and 2.3 minutes. Those are the timescales of inverter duty cycling, of maximum-power-point dither and of cloud-shadow transients: ordinary plant behaviour, not an attack, sitting on frequencies the tokeniser is least able to represent. A monitoring system built on such a forecaster would be systematically less sensitive there, permanently and for reasons no operational log would reveal. That is $T2$, and it is why the candidate set belongs in the deployment documentation of such a system.

The third exposure is governance rather than signal processing. Because the blind set is a function of configuration, a change that no one would route through a security review, a resampling of telemetry from one minute to thirty seconds, a swap of checkpoint, a patch size chosen for latency, relocates it. $H3a$ is what makes this tractable: the relocation is not merely predictable in principle but confirmed to follow f_s/S one for one, so recomputing the set is arithmetic rather than re-measurement.

Controls, residual exposure and conclusion

Two controls in this design are real. Detection has to be placed before tokenisation, on the raw telemetry, because anything placed after it inherits the blind spot it is meant to guard; a spectral monitor on the raw stream, evaluated on the enumerated candidate set, is inexpensive precisely because that set is closed-form. And the semantic layer of Appendix A degrades actuation rather than failing silently: an assessment that is merely PartiallyObservable reduces the battery cap and lowers decision confidence, so uncertainty about the representation reaches the actuator instead of being resolved by assumption. That is the resilience the framework asks for, and it is the part of the design that does not depend on the disputed behavioural claim.

One control is not real, and the report is more useful for having established that. The pre-registered mitigation of Mitigation and Agentic Deployment, increased patch overlap, was refuted: the posterior for δ_O is positive rather than negative in both fitted sets. Even setting that result aside, the arithmetic bounds it, since overlap moves the stride branch and leaves the patch branch untouched. An operator who raises the overlap ratio and records the risk as mitigated has bought a false assurance, which is $T4$ and the reason it is carried at high residual exposure.

The honest conclusion is therefore narrower than the one this section could have asserted and better supported than one it could have assumed. Structural aliasing in a patch-based forecaster is a configuration-determined, attacker-independent weakness whose location is now measured rather than conjectured; its representational cost is established, its propagation to the forecast is not, and the mitigation carried through the design does not close it. The residual exposure is significant, it is concentrated in the patch branch, and it cannot be managed inside the model. It has to be managed around it: by choosing the geometry against the known spectral content of the plant, by monitoring the raw signal upstream of the tokeniser, and by treating the candidate set as a deployment parameter that is recomputed whenever the configuration it derives from changes.

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Appendix A: Ontology of the Photovoltaic Application Domain

This document defines a compact ontology for a photovoltaic system governed by one control agent. It connects environmental and electrical observations to anomaly recognition, Chronos-style patched representations, structural-aliasing evidence, and bounded power-routing decisions [2]. Routing is included only as the operational endpoint of the semantic model; the principal concern remains whether the representation preserves the information on which the agent relies.

Domain Description

The modeled system contains four energy assets: a photovoltaic array, battery storage, an aggregate internal load, and an external grid. The internal load is a passive sink representing a factory or a fleet of machines. Its demand is observed, but it has no agent, machine-level taxonomy, curtailment command, or scheduling model. The external grid is an available bidirectional boundary bus. It supplies residual demand and accepts residual production.

Exactly one `pv:ControlAgent` coordinates the four passive assets. It observes one synchronized control cycle, assigns environmental, anomaly, and observability states, selects a control mode, and emits one or more directional decisions. There is no peer-to-peer negotiation among assets.

The seven admissible transfer directions are

$$\begin{array}{rcc} \text{PV} \to \text{Load}, & \text{PV} \to \text{Battery}, & \text{PV} \to \text{Grid}, \\ \text{Battery} \to \text{Load}, & \text{Battery} \to \text{Grid}, & \text{Grid} \to \text{Load}, \\ & \text{Grid} \to \text{Battery}. & \end{array} \tag{27}$$

They satisfy the cycle-level balance

$$P_{\text{pv}} + P_{\text{bat,dis}} + P_{\text{grid,in}} = P_{\text{load}} + P_{\text{bat,ch}} + P_{\text{grid,out}}. \tag{28}$$

All terms denote non-negative magnitudes; import and export are therefore kept separate rather than encoded by an ambiguous sign.

Purpose and Scope

The ontology provides the smallest semantic layer needed to represent the physical boundary, validated observations, environmental context, three selected anomalies, and the effect of patching on model observability. It does not model individual machines, market prices, dispatch optimisation, grid faults, or a deployed real-time controller. Load demand, battery SoC, asset limits, and the optional grid-exchange reference are asserted or simulated because they are not present in the photovoltaic telemetry dataset.

The environmental states are operational generation proxies, not meteorological diagnoses. Likewise, the routing rules describe a deterministic conceptual policy rather than the central experimental contribution. Structural aliasing remains an empirically tested property of a signal-representation pair; no fixed synthetic frequency is transferred to the one-minute photovoltaic signal.

Ontology Representation

Core classes and closed vocabularies

Table 5 defines the compact class interface. The four asset subclasses form a disjoint union; each `pv:PVSystem` contains exactly one asset of each type and is managed by exactly one control agent. Closed operational vocabularies are represented as named individuals rather than deeper class hierarchies.

Class	Role
`pv:PVSystem`	Aggregate system and scope of each control cycle.
`pv:EnergyAsset`	Parent of the four passive physical assets.
`pv:PhotovoltaicArray`	Intermittent renewable source measured through PV telemetry.
`pv:BatteryStorage`	Bounded storage asset that can charge, discharge, or hold.
`pv:InternalLoad`	Aggregate passive demand of a factory or machine fleet.
`pv:ExternalGrid`	Reliable bidirectional boundary bus for residual balancing.
`pv:ControlAgent`	The single decision-making entity.
`pv:SystemObservation`	Time-stamped snapshot of PV, environment, battery, load, and grid reference.
`pv:TimeSeriesSignal`	Physical signal and its temporal descriptors.
`pv:PatchedRepresentation`	A model input representation defined by patch and stride.
`pv:ObservabilityAssessment`	Evidence linking a signal-representation pair to an observability state.
`pv:GenerationContext`	Operational proxy for expected solar generation.
`pv:AnomalyState`	Detected physical or sensing abnormality.
`pv:ObservabilityState`	Semantic assessment of information retained after patching.
`pv:ControlMode`	Normal, conservative, or safe operating policy.
`pv:PowerRoute`	Closed vocabulary of the seven admissible source–destination pairs.
`pv:PowerRoutingDecision`	One directional transfer or hold decision with an optional bounded setpoint.

Table 5: *Core classes in the simplified ontology.*

The closed vocabularies are:

Generation context: ClearSky, PartialCloud, Overcast, and Nighttime.
Anomaly: InverterTrip, RampEvent, and SensorMalfunction.
Observability: Observable, PartiallyObservable, and StructurallyAliased.
Control mode: NormalMode, ConservativeMode, and SafeMode.
Power route: PVToInternalLoad, PVToBattery, and PVToExternalGrid;
BatteryToInternalLoad and BatteryToExternalGrid;
ExternalGridToInternalLoad and ExternalGridToBattery.

Formally, each vocabulary class is an OWL-style enumeration of the listed named individuals (`oneOf`). The five vocabulary classes are pairwise disjoint, and all listed values are declared globally different (`AllDifferent`). This prevents a value such as SafeMode from being silently identified with ClearSky or NormalMode under open-world semantics. Each valid time-stamped observation has exactly one generation context. Each completed observability assessment has exactly one observability state, and each routing decision records exactly one control mode. Co-issued decisions based on the same synchronized observation must record the same mode. Anomalies are not declared mutually exclusive because, for example, an abrupt inverter trip also creates a large power change. Deterministic control is obtained through the priority rules in the closed-loop section.

Object properties

The semantic relations in Table 6 connect the physical boundary to the inference and control layers. Each active `pv:PowerRoutingDecision` has exactly one route value, one source, and one distinct destination. Its non-negative setpoint magnitude is optional because `Balance` may denote passive residual exchange. A cycle that simultaneously serves the load, charges the battery, and exports a residual uses three instances of the same decision class, all based on the same observation and mode.

Property	**Domain**	**Range or meaning**
`pv:hasAsset`	`pv:PVSystem`	`pv:EnergyAsset`
`pv:managedBy`	`pv:PVSystem`	exactly one `pv:ControlAgent`
`pv:hasObservation`	`pv:PVSystem`	`pv:SystemObservation`
`pv:hasSignal`	`pv:PhotovoltaicArray`	`pv:TimeSeriesSignal`
`pv:hasRepresentation`	`pv:TimeSeriesSignal`	`pv:PatchedRepresentation`
`pv:assessesRepresentation`	`pv:ObservabilityAssessment`	exactly one `pv:PatchedRepresentation`
`pv:hasGenerationContext`	`pv:SystemObservation`	zero or one `pv:GenerationContext` value
`pv:hasAnomaly`	`pv:SystemObservation`	zero or more `pv:AnomalyState` values
`pv:hasObservabilityState`	`pv:ObservabilityAssessment`	one `pv:ObservabilityState` value
`pv:selectsMode`	`pv:PowerRoutingDecision`	exactly one `pv:ControlMode` value
`pv:makesDecision`	`pv:ControlAgent`	`pv:PowerRoutingDecision`
`pv:basedOn`	`pv:PowerRoutingDecision`	observation, assessment, or inferred state
`pv:hasRoute`	`pv:PowerRoutingDecision`	zero or one `pv:PowerRoute` value
`pv:hasSourceAsset`	`pv:PowerRoutingDecision`	zero or one `pv:EnergyAsset`
`pv:hasDestinationAsset`	`pv:PowerRoutingDecision`	zero or one `pv:EnergyAsset`

Table 6: *Object-property signatures.*

`pv:hasRepresentation` is intentionally non-functional because one signal can be evaluated under several (P, S) configurations. Each assessment identifies exactly which representation it evaluates. Each decision is based on exactly one current observation and its completed assessment; its own `pv:selectsMode` value therefore scopes the mode without accumulating incompatible modes on the persistent agent. The range of `pv:basedOn` is the union of observation and inferred-assessment entities, rather than the intersection that multiple RDFS range declarations would imply.

OWL's open-world semantics do not by themselves validate that endpoint types match a named route. The operational validator therefore admits only the seven ordered pairs in Equation 27, rejects identical endpoints, and checks consistency among `pv:hasRoute`, source, and destination. A hold decision has no route or endpoints and has a zero or absent setpoint. The `pv:basedOn` relation is retained only as the minimum evidence link required to justify a selected mode; no separate trace model is introduced.

Data properties and units

The synchronized observation combines the measured photovoltaic and environmental variables with asserted or simulated battery, load, and grid-reference values. The control-critical quantities are PV power P_{pv}, tilted irradiance G_{tilt}, normalized battery state of charge $s \in [0, 1]$, and aggregate load demand $P_{\mathrm{load}} \geq 0$. Each has a separate validity flag derived from presence, finiteness, freshness, and a configured physical range. Horizontal irradiance, air temperature, wind speed, and wind direction remain environmental evidence but do not independently block the controller. The signed `pv:gridExchangeReferenceW` is optional, with positive values denoting export; it is usable only when present, finite, and fresh.

Missing or stale data cannot be inferred safely from OWL's open-world assumption alone. An operational validation layer therefore compares the observation time stamp with a configurable staleness limit, validates numeric ranges, and asserts the four critical Boolean flags consumed by the ontology. This keeps data-quality validation distinct from semantic classification.

Owner	Data properties
`pv:SystemObservation`	`pv:pvPowerW`, `pv:powerRampRateWPerS`, `pv:loadDemandW`, optional `pv:gridExchangeReferenceW`, `pv:tiltedIrradianceWm2`, `pv:horizontalIrradianceWm2`, `pv:airTemperatureC`, `pv:windSpeedMs`, `pv:windDirectionDeg`, `pv:batterySoC`, `pv:observationTimestamp`, `pv:pvPowerValid`, `pv:tiltedIrradianceValid`, `pv:batterySoCValid`, and `pv:loadDemandValid`.
`pv:BatteryStorage`	`pv:capacityWh`, `pv:reserveSoC`, `pv:targetSoC`, `pv:maximumSoC`, `pv:maximumChargePowerW`, and `pv:maximumDischargePowerW`.
`pv:TimeSeriesSignal`	`pv:signalFrequencyHz`, `pv:samplingFrequencyHz`, `pv:periodSamples`, `pv:amplitudeW`, `pv:timeSpanSeconds`, and `pv:phaseOffsetSamples`.
`pv:PatchedRepresentation`	`pv:patchSize`, `pv:stride`, and derived `pv:overlapRatio`.
`pv:ObservabilityAssessment`	`pv:analysisStatus`, `pv:assessmentMethod`, `pv:assessmentMetric`, `pv:strideLockingRatio`, `pv:patchLockingRatio`, `pv:lockingTolerance`, `pv:effectEstimate`, `pv:posteriorLossProbability`, `pv:credibleIntervalLower`, `pv:credibleIntervalUpper`, and `pv:assessmentTimestamp`.
`pv:PowerRoutingDecision`	optional non-negative `pv:powerSetpointW`, `pv:decisionConfidence`, `pv:batteryCommand` in {Charge, Discharge, Hold}, `pv:pvCommand` in {Connect, Isolate}, and `pv:gridCommand` in {Import, Export, Balance}.

Table 7: *Quantitative property groups and units encoded by their names.*

A valid patched representation satisfies $P \geq 1$ and $1 \leq S \leq P$, and stores the overlap ratio O of Equation 20 as descriptive metadata. It is not used as a standalone rule for declaring aliasing.

Inference Rules

Environmental generation context

Let $g_0 < g_1 < g_2$ be site-configurable tilted-irradiance thresholds and let G_t denote the validated value of G_{tilt} at time t. The active generation context is

$$C_G(G_t) = \begin{cases} \textsf{Nighttime}, & G_t \leq g_0, \\ \textsf{Overcast}, & g_0 < G_t < g_1, \\ \textsf{PartialCloud}, & g_1 \leq G_t < g_2, \\ \textsf{ClearSky}, & G_t \geq g_2. \end{cases} \tag{29}$$

Values $g_0 = 0$, $g_1 = 150$, and $g_2 = 800\ \mathrm{W\,m^{-2}}$ are retained only as an illustrative configuration compatible with the current coursework. They are not universal meteorological boundaries. In particular, low irradiance can also be caused by solar position; the four values must therefore be read as generation-context proxies rather than certain weather diagnoses.

Anomaly detection and precedence

Let $v_P(t)$, $v_G(t)$, $v_s(t)$, and $v_L(t)$ denote the validity of PV power, tilted irradiance, battery SoC, and aggregate load demand. The control-critical validity predicate is

$$V_t = v_P(t) \wedge v_G(t) \wedge v_s(t) \wedge v_L(t). \tag{30}$$

If V_t is false, no physical anomaly or forecast-driven route is inferred from that observation. A valid irradiance value may still support a descriptive generation context, but the operational validator asserts SensorMalfunction and the cycle proceeds directly to safe-mode selection. Within a validated cycle, absence of the three listed anomaly values is interpreted locally as "none detected"; this is a scoped closed-world policy, not a global OWL assumption. The signed ramp rate, measured in watts per second when Δt is expressed in seconds, is

$$r_t = \frac{P_t - P_{t-1}}{\Delta t}. \tag{31}$$

The anomaly rules are expressed with configurable thresholds: g_{trip} for sufficient irradiance, p_0 for near-zero power, and $r_{\max}$ for maximum admissible ramp magnitude.

$$\neg V_t \longrightarrow \mathsf{SensorMalfunction}, \tag{32}$$
$$V_t \wedge G_t \geq g_{\text{trip}} \wedge |P_t| \leq p_0 \longrightarrow \mathsf{InverterTrip}, \tag{33}$$
$$V_t \wedge V_{t-1} \wedge |r_t| > r_{\max} \longrightarrow \mathsf{RampEvent}. \tag{34}$$

No numerical value is inherited for $r_{\max}$ because it must be calibrated after invalid values and outliers have been excluded. The controller evaluates sensor validity first. A sensor malfunction therefore cannot be reinterpreted as a physical ramp or inverter trip. If a valid inverter trip also satisfies the ramp predicate, the safe response to the trip takes precedence over ramp smoothing.

Signal observability and structural aliasing

For a signal frequency f, sampling frequency f_s, patch size P, and stride S, define a stride-lock ratio and a patch-periodicity ratio

$$\rho_S = \frac{fS}{f_s}, \qquad \rho_P = \frac{fP}{f_s}. \tag{35}$$

A stride phase-lock candidate L_S and a within-patch repetition candidate L_P are

$$\begin{aligned} L_S &= \left[\exists m \in \mathbb{Z}^+ : |\rho_S - m| \leq \varepsilon_S\right], \\ L_P &= \left[\exists k \in \mathbb{Z}^+ : |\rho_P - k| \leq \varepsilon_P\right], \\ C &= (L_S \vee L_P) \wedge \left(0 < f < \frac{f_s}{2}\right), \end{aligned} \tag{36}$$

where ε_S and ε_P are tied to frequency resolution and are never hardcoded frequency bands. Only L_S expresses equality of consecutive patch phases; L_P records whole-cycle repetition inside a patch and is retained as an H3 experimental factor. Their union Z is a candidate geometry class, not proof that two domain concepts are indistinguishable.

This is a division of labour, and it bounds what the ontology can conclude on its own. Phase-locking is *derivable*: L_S, L_P and Z follow by rule from `pv:signalFrequencyHz`, `pv:samplingFrequencyHz`, `pv:patchSize` and `pv:stride`, all of which the knowledge base already holds, so a reasoner can enumerate the candidate set for any configuration without observing a model. Structural aliasing is *not* derivable and is only recorded: it enters through Y below, which is posterior evidence from an experiment external to the ontology, and no rule here can produce it. An agent using this ontology therefore detects phase-locking and reports structural aliasing.

Let q denote the task-relevant information being tested, for example frequency or amplitude. Let $\Delta_{\text{loss}}(q)$ be its predeclared behavioral or representational loss relative to matched controls, δ the maximum practically negligible loss, D the experimental evidence, and $\tau > 1/2$ the posterior confidence threshold. Define confirmed information loss Y and confirmed preservation R as

$$Y = \left[\Pr(\Delta_{\text{loss}}(q) > \delta \mid D) \geq \tau\right], \qquad R = \left[\Pr(\Delta_{\text{loss}}(q) \leq \delta \mid D) \geq \tau\right]. \tag{37}$$

Because $\tau > 1/2$, Y and R cannot both hold for the same assessment. The assessment interface receives the offline evidence produced by the Bayesian analysis of this report: a local amplitude-recovery contrast with a heavy-tailed Student-t_4 likelihood and a Gamma regression with log link for non-negative prequential-MDL codelength. The population contrast uses the weakly informative prior $\bar{\beta} \sim \text{Student-}t_4(0, 0.5)$; the Gamma model uses $\theta_{\text{lock}} \sim \mathcal{N}(0, 0.5)$ and shape $k \sim \text{Gamma}(2, 0.1)$. The stored record identifies the tested (P, S) configuration, matched controls, method and metric, effect estimate, 95% credible interval, and posterior probability of at least 20% attenuation. These are evidence fields, not universal ontology thresholds. Unless that inference satisfies the predeclared criterion, no instance may be asserted as StructurallyAliased.

The three observability values are assigned as follows:

$$C_O = \begin{cases} \mathsf{StructurallyAliased}, & X, \\ \mathsf{Observable}, & \neg X \wedge R, \\ \mathsf{PartiallyObservable}, & \neg X \wedge \neg R, \end{cases} \qquad X = Z \wedge Y. \tag{38}$$

Observable therefore means that positive evidence shows the tested representation preserves the task-relevant information within tolerance δ. It does not mean global observability of the physical system, nor does it merely mean that aliasing was not found. Candidate geometry may still be Observable when the experiment demonstrates preservation. PartiallyObservable is the epistemic fallback: preservation has not been certified, while structural aliasing has not been confirmed. It includes a candidate with inconclusive evidence, an unfinished test, a credible interval crossing the decision threshold, conflicting metrics, or no preservation evidence. The term means *partially trusted*, not that a fixed fraction of samples or sensors is visible.

H1 compares each candidate frequency with matched nearby non-candidate controls and predicts higher MDL codelength or lower amplitude recovery; absent or non-localized loss refutes H1. H2 tests whether the loss at a candidate frequency remains approximately stable across sampled phase offsets; systematic phase dependence refutes that hypothesis. H3 tests whether candidate degradation sites move proportionally to $1/S$ or $1/P$ when the corresponding parameter varies. These hypotheses contribute controlled evidence to D, but remain falsifiable experimental factors rather than ontology subclasses or automatic action rules. Sensor validity is modeled separately as an anomaly and never relabels a model representation as structurally aliased.

When an experimental generator provides a phase ϕ in radians, the corresponding temporal offset is converted before storage:

$$o_{\text{samples}} = \frac{\phi f_s}{2\pi f}. \tag{39}$$

Neither ϕ nor o_{samples} is compared directly with stride to infer an anomaly.

The semantic propagation path is explicit:

$$\begin{aligned} \texttt{pv:TimeSeriesSignal} &\longrightarrow \texttt{pv:PatchedRepresentation} \\ &\longrightarrow \texttt{pv:ObservabilityAssessment} \\ &\longrightarrow \texttt{pv:ControlMode} \longrightarrow \texttt{pv:PowerRoutingDecision}. \end{aligned} \tag{40}$$

Aliasing can therefore reduce confidence in forecast-derived production, ramp expectations, forecast-driven anomaly recognition, and routing. It does not alter the directly observed values of G_{tilt}, P_{pv}, load demand, or SoC; those values are governed by the separate validity path.

Closed-Loop Control

Mode selection

The agent evaluates each synchronized control cycle using the strict priority

$$\mathsf{SafeMode} \succ \mathsf{ConservativeMode} \succ \text{RampOverride} \succ \text{NormalRouting}. \tag{41}$$

Let N_{safe} mean that SensorMalfunction or InverterTrip is present. The selected mode is

$$M_t = \begin{cases} \mathsf{SafeMode}, & N_{\text{safe}} \vee (C_O = \mathsf{StructurallyAliased}), \\ \mathsf{ConservativeMode}, & C_O = \mathsf{PartiallyObservable}, \\ \mathsf{NormalMode}, & \text{otherwise}. \end{cases} \tag{42}$$

In SafeMode, the battery command is Hold and the grid balances the internal load. The PV command is Isolate only for an inverter trip; for sensor malfunction or confirmed structural aliasing it remains connected, but forecast- and ramp-driven battery decisions are suspended. When load demand itself is invalid, Balance carries no invented numerical setpoint. In ConservativeMode, the same seven route types remain available with a configurable reduced battery-power cap, normal target/reserve bounds, and low decision confidence. A concurrent ramp cannot activate either extended ramp bound; this implements the precedence in Equation 41.

Routing policy

The ordinary policy serves the internal load before charging the battery or exporting energy. Let

$$P_{\text{sur}} = (P_{\text{pv}} - P_{\text{load}})^{+}, \qquad P_{\text{def}} = (P_{\text{load}} - P_{\text{pv}})^{+}. \tag{43}$$

Here *daylight* abbreviates {ClearSky, PartialCloud, Overcast}. A cycle issues a route only for a strictly positive transfer and may issue several co-mode decisions. All setpoints are clipped to SoC, capacity, and charge/discharge limits.

Figure 5 shows the priority scan and Table 8 states its boundary cases. The conservative branch rejoins the ordinary load-first router under its reduced cap. The ramp branch is available only in NormalMode.

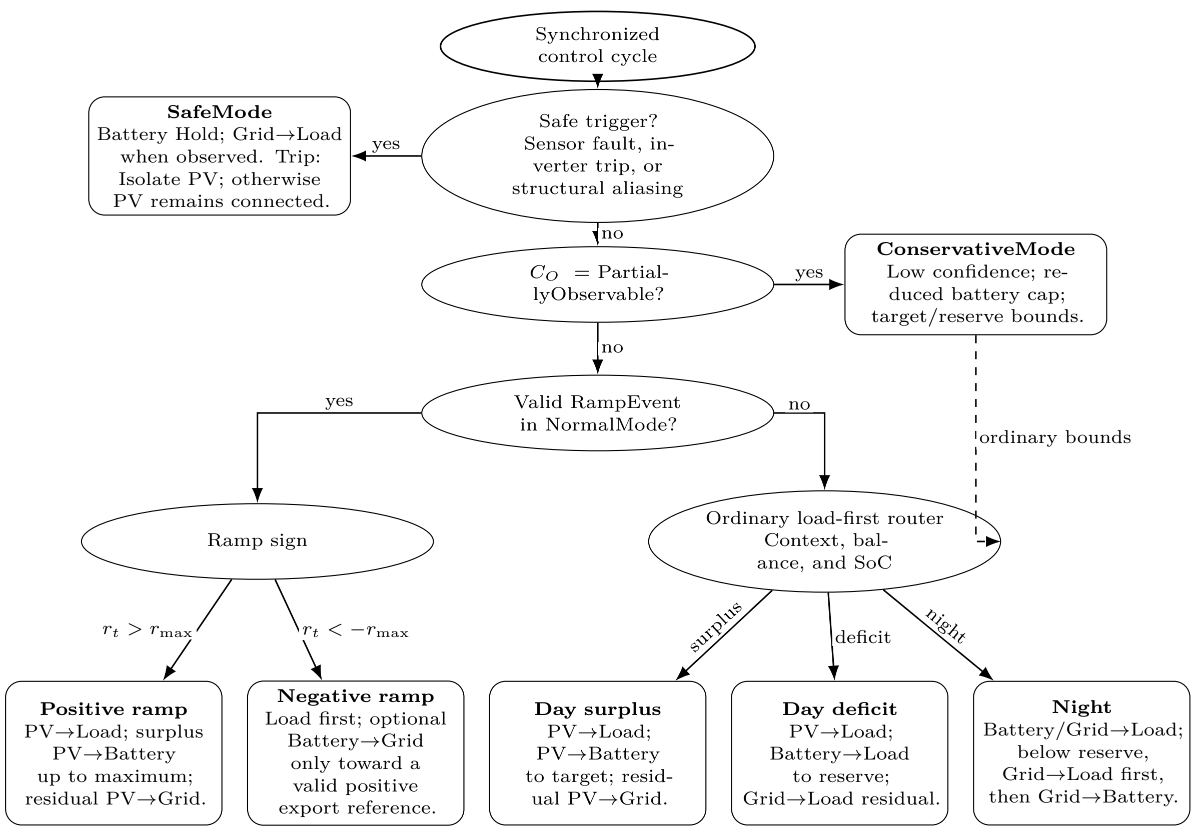

Figure 5: *Load-first routing decision tree. Elliptical bubbles are ordered gates; rounded bubbles are terminal policies. Safe and conservative decisions precede the normal-mode ramp override, while the ordinary router covers daylight surplus, daylight deficit, and nighttime operation.*

Condition	Ordered route set	Constraint
Daylight surplus	PV→Load; PV→Battery; PV→Grid	Serve demand, charge to target, then export the residual.
Daylight deficit	PV→Load; Battery→Load; Grid→Load	Discharge only to reserve; the grid covers the remaining deficit.
Night, $s > s_{\text{reserve}}$	Battery→Load; Grid→Load	Battery serves demand to reserve; the grid supplies the residual.
Night, $s \leq s_{\text{reserve}}$	Grid→Load; if $s < s_{\text{reserve}}$, Grid→Battery	Supply the load first; charge only below reserve and stop at reserve.
Positive ramp with surplus	PV→Load; PV→Battery; PV→Grid	NormalMode may charge beyond target, never beyond maximum SoC.
Negative ramp with valid positive reference	load-serving routes; Battery→Grid	Export only after serving the load, within residual discharge capacity and above reserve.
Negative ramp without usable reference	ordinary surplus/deficit routes	Missing, stale, non-finite, zero, or negative reference disables export catch-up.
ConservativeMode	ordinary load-first routes	Reduced battery cap; target/reserve bounds; no ramp extensions.
SafeMode	Grid→Load when observed	Battery Hold; grid Balance; isolate PV only for inverter trip.

Table 8: *Routing matrix for the seven admissible transfer directions.*

For a negative ramp, Battery→Grid is admissible only after the internal load has been fully served, the export reference is present, fresh, finite, and positive, measured export is below that reference, the battery remains above reserve, and discharge capacity remains after load support. If any condition fails, the ordinary load-first route applies. At night, Grid→Battery begins only after Grid→Load has been satisfied and stops when reserve is restored. These rules preserve Equation 28 without turning the routing policy into a separate optimisation problem.

Appendix B: Light TSMixup: Description and Pseudocode

Light TSMixup is a controlled probing variant of the TSMixup augmentation described in the Chronos data-generation procedure[2]. The original algorithm samples subsequences from a collection of real training datasets, applies mean scaling to each sampled series, draws mixing coefficients from a symmetric Dirichlet distribution and returns their convex combination.

The variant used here preserves that stochastic mixing structure and replaces the real dataset collection with a controlled pool of sinusoidal sources, each defined by its frequency, amplitude, phase, source length and sampling frequency. The consequence is that every generated series remains a known superposition of known components: a tone injected into it can still be identified exactly, and any loss measured at a predicted site cannot be attributed to unknown spectral content in the background. This is what makes it usable as a probe rather than only as an augmentation.

Four parameters control what the generator produces. The maximum number of components K bounds how many sinusoids may enter one mixture, and therefore how spectrally dense the background can be. The Dirichlet concentration α sets how evenly the mixing weights are spread: values near zero concentrate almost all the mass on one component and return something close to a single sinusoid, while larger values return balanced mixtures. The augmented length is drawn between $l_{\min}$ and $l_{\max}$, and the sampling frequency f_s fixes the mapping between the source frequencies and the sample grid, so it is what ties the pool to the lock arithmetic of Phase Locking. Algorithm 1 states the procedure together with the values adopted.

Algorithm 1 Light TSMixup: Controlled Sinusoidal Time Series Mixup

Require: Sinusoidal pool $\mathcal{P} = \{p_1, \ldots, p_{N_p}\}$, where each p_n is defined by frequency f_n, amplitude A_n, phase ϕ_n, source length T_n, and sampling frequency f_s; maximum components $K = 10$; symmetric Dirichlet concentration $\alpha = 1.5$; minimum and maximum augmented lengths ($l_{\min} = 128, l_{\max} = 2048$).

Ensure: An augmented time series.

```
k ~ U{1, K}                                          ▷ number of sinusoidal components to mix
l ~ U{l_min, l_max}                                  ▷ length of the augmented time series
for i ← 1 to k do
    n ~ U{1, N_p}                                    ▷ sample a sinusoidal source
    x^(i)_{1:l} ← SampleSegment(p_n, l, f_s)         ▷ generate or crop a length-l segment
    x̃^(i)_{1:l} ← x^(i)_{1:l} / ( (1/l) Σ_{j=1}^{l} |x^(i)_j| )   ▷ apply mean scaling
end for
[λ_1, ..., λ_k] ~ Dir([α_1 = α, ..., α_k = α])       ▷ sample mixing weights
return Σ_{i=1}^{k} λ_i x̃^(i)_{1:l}                   ▷ take a weighted combination of scaled signals
```

Where: $k \sim U\{1, K\}$; $l \sim U\{l_{\min}, l_{\max}\}$; $n \sim U\{1, N_p\}$; $x_{1:l}^{(i)} \leftarrow \mathrm{SampleSegment}(p_n, l, f_s)$; $\tilde{x}_{1:l}^{(i)} \leftarrow x_{1:l}^{(i)} \Big/ \left(\frac{1}{l} \sum_{j=1}^{l} \left| x_j^{(i)} \right| \right)$; $[\lambda_1, \ldots, \lambda_k] \sim \mathrm{Dir}([\alpha_1 = \alpha, \ldots, \alpha_k = \alpha])$; return $\sum_{i=1}^{k} \lambda_i \tilde{x}_{1:l}^{(i)}$

Appendix C: KernelSynth: Description, Pseudocode and Kernel Bank

KernelSynth is the Gaussian-process generator of the same Chronos data-generation procedure[2]. Where Light TSMixup can only produce superpositions of sinusoids, KernelSynth composes covariance functions and samples from the resulting prior, so its output carries trends, non-stationary structure and broadband content as well as periodicity. It is included precisely because a phenomenon that appears only on tones and simple mixtures would be an artefact of the probe; a phenomenon that survives on signals resembling the corpus the model was trained on is a property of the model.

The procedure draws j kernels with replacement from a fixed bank $\mathcal{K}$, composes them pairwise under a binary operator sampled between sum and product, and draws one realisation from the zero-mean Gaussian process with the composite covariance. Sum composition superposes two behaviours, so a periodic kernel added to a linear one produces an oscillation riding on a trend; product composition modulates one by the other, so the same pair multiplied produces an oscillation whose amplitude grows along the series. That distinction is what gives the family its range from a single draw of the same bank.

Three parameters control the output. The maximum number of kernels J bounds the compositional depth, and therefore how far a draw can depart from the behaviour of a single kernel. The generated length l_{syn} fixes the support of the covariance, and enters the periodic kernel directly through the ratio p/l_{syn}, so the period values of Table 9 are expressed relative to the length of the series rather than in samples. The jitter ϵ is a numerical term added to the diagonal to keep the covariance positive definite, and is set to zero unless a draw fails to factorise. Algorithm 2 states the procedure and Table 9 the bank, with the hyperparameter values each kernel is drawn from: the length scale l of the RBF kernel and the shape α of the rational quadratic control how quickly the process decorrelates, the noise scale σ_n how much unstructured content is added, and the period set of the periodic kernel spans the daily, weekly, monthly and yearly cycles of the original recipe.

Algorithm 2 KernelSynth: Synthetic Data Generation using Gaussian Processes

Require: Kernel bank $\mathcal{K}$ listed in Table 9; maximum kernels per time series $J = 5$; generated length $l_{\text{syn}} = 1024$; optional numerical jitter $\epsilon = 0$.

Ensure: A synthetic time series $x_{1:l_{\text{syn}}}$.

$j \sim U\{1, J\}$ ▷ sample the number of kernels
$\{\kappa_1(t,t'), \ldots, \kappa_j(t,t')\} \overset{\text{i.i.d.}}{\sim} \mathcal{K}$ ▷ sample kernels with replacement
$t \leftarrow \text{linspace}(0, 1, l_{\text{syn}})$
$\kappa^\star(t,t') \leftarrow \kappa_1(t,t')$ ▷ initialize the composite covariance
for $i \leftarrow 2$ to j **do**
    $\star \sim U\{+, \times\}$ ▷ sample a binary composition operator
    $\kappa^\star(t,t') \leftarrow \kappa^\star(t,t') \star \kappa_i(t,t')$ ▷ compose kernels
end for
if $\epsilon > 0$ **then**
    $\kappa^\star(t,t') \leftarrow \kappa^\star(t,t') + \epsilon I$
end if
$x_{1:l_{\text{syn}}} \sim \mathcal{GP}(0, \kappa^\star(t,t'))$ ▷ sample from the GP prior
return $x_{1:l_{\text{syn}}}$

Table 9: *Kernel bank $\mathcal{K}$ used by KernelSynth.*

Kernel	Formula	Hyperparameters
Constant	$\kappa_{\text{Const}}(x, x') = C$	$C = 1$
White Noise	$\kappa_{\text{White}}(x, x') = \sigma_n \cdot \mathbf{1}_{x=x'}$	$\sigma_n \in \{0.1, 1\}$
Linear	$\kappa_{\text{Lin}}(x, x') = \sigma_0^2 + x \cdot x'$	$\sigma_0 \in \{0, 1, 10\}$
RBF	$\kappa_{\text{RBF}}(x, x') = \exp\left(-\frac{\|x-x'\|^2}{2l^2}\right)$	$l \in \{0.1, 1, 10\}$
Rational Quadratic	$\kappa_{\text{RQ}}(x, x') = \left(1 + \frac{\|x-x'\|^2}{2\alpha}\right)^{-\alpha}$	$\alpha \in \{0.1, 1, 10\}$, $l = 1$
Periodic	$\kappa_{\text{Per}}(x, x') = \exp\left(-2\sin^2\left(\pi \frac{\|x-x'\|}{p/l_{\text{syn}}}\right)\right)$	$p \in \{24, 48, 96, 168, 336, 672,$ $7, 14, 30, 60, 365, 730,$ $4, 26, 52, 4, 6, 12, 4, 40, 10\}$

Model	**Claim**	**Estimand**	**Predicts**	**Input**
A	$H1$ behavioural	$e^{\bar{\beta}}$: recovery at a lock / at its controls	< 1	`contrasts`
B	$H1$ representational	$e^{\theta_{\text{lock}}}$: codelength at a lock / elsewhere	> 1	`mdl_cells`
C	$H2$	σ_ϕ: spread of the deficit across phase	≈ 0	`contrasts`
D1	$H3$ location	θ_S, θ_P: dip depth on each grid, by LOO	both < 0	`collapse`
D2	$H3a$ stride branch	κ_S: measured / stride-predicted spacing	$= 1$	`sites`
D2	$H3b$ patch branch	κ_P: measured / patch-predicted spacing	$= 1$	`sites`
A′	$M1$ mitigation	δ_O: change in the deficit per unit of overlap	< 0	`contrasts`

Table 10: *The models, the quantity whose posterior decides each claim, the value the claim predicts, and the probing table each is fitted to. $H1$ is tested twice, on the forecast (A) and on the representation (B), so that a discrepancy between them stays visible. $H3$ is split into its two branches, since $\mathcal{F}_{lock}$ is a union and a frequency may belong to either. $M1$ is not a separate model but the configuration level of A, promoted to a named estimand.*

Appendix D: Bayesian Model Specifications

This appendix states in full the five models summarised in Bayesian Analysis. Table 10 lists them side by side; the sections that follow give, for each, what it measures, how the measurement is constructed, and what every parameter of it means. The equations are repeated here so that the appendix can be read on its own.

Sampling and diagnostics

Every model is fitted by the No-U-Turn sampler with the same settings, so that none receives a more or less careful exploration than another: four independent chains, 2000 warm-up draws per chain used to adapt the step size and discarded, 2000 retained draws per chain, and a target acceptance rate of 0.9, which trades a shorter step for a more cautious traversal of the posterior geometry. Group effects are written non-centred, $u = \sigma_u z$ with $z \sim \mathcal{N}(0, 1)$: the two forms are the same distribution, but the first removes the funnel that produces divergences when a group scale is small, which is exactly what is expected if $H2$ holds.

Three diagnostics are preregistered and gate interpretation. $\hat{R} < 1.01$ measures agreement between the four chains, and values near one indicate that each has converged to the same distribution. Bulk and tail effective sample size above 1000 counts the genuinely independent draws remaining after correcting for autocorrelation, the tail figure guarding the interval endpoints that the credible intervals are read from. Zero divergences requires that the sampler encountered no geometric pathology it could not traverse. A fit failing any of the three is not used to support a claim, although it may still be reported.

Where two formulations compete they are compared by leave-one-out cross-validation. The quantity read is the difference in expected log pointwise predictive density, elpd_diff, with its standard error dse; a difference smaller than twice its standard error is reported as inconclusive rather than rounded into a decision, so that a nominal ranking is not mistaken for evidence.

Model A, the behavioural reading of $H1$

Model A is fitted to the paired contrast of Equation 21,

$$d = \log(R_k + 0.01) - \tfrac{1}{2}\left[\log(R_- + 0.01) + \log(R_+ + 0.01)\right], \tag{44}$$

where $R = A_{\text{pred}}/A_{\text{true}}$ is the amplitude of the tone in the median forecast divided by its amplitude in the true continuation, both fitted by least squares over the same horizon so that the ratio is one of like for like. The additive constant 0.01 keeps the logarithm finite where a tone is not rebuilt at all. The three members of a triplet share a background realisation and a phase, so a difference between them cannot be produced by the background or by the alignment of the window; only the frequency changes. The contrast is negative when the candidate recovers less than its neighbours, which is the direction $H1$ predicts.

The fit is

$$d_i \sim \text{Student-}t_4(\mu_i,\ \sigma), \quad \mu_i = \beta_{c[i]} + u_{k[i]} + u_{b[i]}, \quad \beta_c \sim \mathcal{N}\Big(\bar{\beta} + \delta_O \widetilde{O}_c + \delta_P \widetilde{\log P}_c,\ \tau\Big), \tag{45}$$

and has four components, each motivated separately.

The linear predictor carries three offsets, one per grouping the observation belongs to: β_c per configuration, u_k per lock harmonic and u_b per background draw, so that neither an unusual frequency nor a single corpus can be mistaken for the effect. The scale σ_b of the background offset is what separates generator-specific variation from the phenomenon itself: a recovery difference between Light TSMixup and KernelSynth backgrounds is attributed to the generator rather than inflating or deflating the estimate of $\bar{\beta}$.

The likelihood is Student-t_4 rather than Gaussian, and the choice is not stylistic. Where the model rebuilds nothing, recovery is near zero and the logarithm of Equation 44 returns very large contrasts. A Gaussian charges a residual quadratically, so a handful of such rows would determine the estimate and the fit would contort itself to accommodate them; the t_4 charges the same residual logarithmically, so an outlier is tolerated rather than obeyed, and a few dead frequencies cannot set $\bar{\beta}$.

The configuration effects β_c are in turn drawn around a population mean, which makes the geometries instances of one phenomenon rather than unrelated experiments. Two consequences follow. The first is that $\bar{\beta}$ exists as a parameter at all, which $H1$ requires, being a claim about patch-based tokenisation and not about a single checkpoint. The second is that each geometry contributes equal weight, so the largest design cannot dominate the estimate when the number of usable triplets varies several-fold with the in-band lock count.

Two configuration-level covariates enter that population mean, and they are estimable together rather than confounded because the design varies overlap and patch size independently. Overlap enters as a ratio and patch size as $\log P$, because the patch grid has spacing f_s/P and equal steps in $\log P$ are equal ratios of spacing. The tilde denotes centring on the design, $\widetilde{O}_c = (O_c - \bar{O})/0.5$ and $\widetilde{\log P}_c = \log P_c - \overline{\log P}$, which leaves $\bar{\beta}$ the effect at the average configuration rather than an extrapolation to $O = 0$ and $P = 1$.

The estimand is $\bar{\beta}$, reported as $e^{\bar{\beta}}$: the multiplicative change in forecast amplitude recovery at a candidate frequency relative to its controls. The overlap slope δ_O is not a separate model but this same configuration level promoted to a named estimand, and it carries $M1$.

Model B, the representational reading of $H1$

Model B replaces the forecast with a measurement of the internal state. At each candidate frequency f_c, examples are drawn at $f_c - 1\,\text{Hz}$ and $f_c + 1\,\text{Hz}$, the representation is captured, and a linear probe is charged the prequential description length of the labels[9]: the data are split into successive chunks, each chunk encoded by a probe trained only on the chunks before it, and the cost accumulated in bits. A probe that memorises noise pays for itself and gains nothing, which is why description length is preferred to accuracy. The reading is direct: at a frequency whose neighbourhood has become linearly indistinguishable in representation space, the labels cost more bits.

Ten states are probed per geometry: the encoder [REG] token after each of the four encoder blocks, the decoder state after each of the four decoder blocks, the pre-projection vector and the output quantile head. The decoder-side vectors are named decoder states rather than [REG] tokens because Chronos-Bolt carries [REG] in the encoder only and the decoder is entered through a single start token.

The fit is

$$L_i \sim \mathrm{Gamma}(r,\ r/\mu_i)\,, \qquad \log \mu_i = \alpha_0 + \theta_{\mathrm{lock}}\,\mathrm{IsLocked}_i + s_{\mathrm{st}[i]} + g_{\mathrm{geo}[i]}. \tag{46}$$

A codelength is strictly positive, right-skewed and has a variance that grows with its mean, which is what the Gamma likelihood with a log link is for; r is its shape and carries the dispersion. The log link makes θ_{lock} multiplicative, so $e^{\theta_{\mathrm{lock}}}$ is the factor by which the cost of the labels expands at a candidate frequency. The offsets s_{st} and g_{geo} are zero-mean and absorb the level differences between probe stages and between geometries, which run to several-fold and would otherwise be attributed to the lock indicator. The unadjusted form, without those offsets, is fitted alongside and compared by LOO, so the blocking decision is reported rather than assumed.

The scope of the probe task is deliberately narrow. The seven hierarchical band tasks used in the exploratory sweep of Signal Generation, Experimental Setup and Workflow each summarise hundreds of frequencies in one codelength, so the lock indicator has nothing to attach to; they are retained as a descriptive cross-check of overall decodability and are not fitted by any Bayesian model.

Model C, $H2$

Model C is Model A on the sign-reversed, per-phase response $y_i = \log(R_{\mathrm{ctrl},i} + 0.01) - \log(R_{\mathrm{lock},i} + 0.01)$, positive when the candidate is worse. The configuration-level covariates are dropped, because $H2$ is not a question about the design axes, and the phase circle is cut into eight equal slices, each given its own offset:

$$y_i \sim \text{Student-}t_4(\mu_i,\ \sigma), \qquad \mu_i = \beta_{c[i]} + u_{k[i]} + u_{b[i]} + u_{p[i]}, \qquad u_p \sim \mathcal{N}(0, \sigma_\phi). \tag{47}$$

The estimand σ_ϕ is the spread of the eight offsets and reduces the hypothesis to one number. Slices are preferred to a fitted sinusoidal or polynomial term because they assume nothing about the shape a phase dependence might take: any systematic pattern across the circle, of whatever form, widens the spread. Two readings are reported and can disagree, which is itself informative. The magnitude reading is $\Pr(\sigma_\phi < \log 1.1)$, whether phase dependence is practically negligible; the evidential reading is the LOO comparison against the same model with u_p removed, whether allowing phase to matter predicts held-out data better. A small but consistently detected dependence would show a low equivalence probability together with a LOO preference for the phase model.

Both Model A and Model C are written non-centred, $u = \sigma_u z$ with $z \sim \mathcal{N}(0, 1)$, which is the same distribution but gives the sampler a geometry without the funnel that appears when a group scale is small — which is exactly what is expected here if $H2$ holds.

Model D1, the location of the sites

The measurement is the token collapse $z_g(f)$: the input patches of one context are embedded, and the standard deviation of those embeddings across patches is averaged over the embedding dimensions. It is exactly zero when consecutive patches embed to the same vector, which is the degeneracy the stride condition predicts, and it is read before any transformer block, so it measures the tokenisation and not what the network subsequently does with it. Each curve is normalised by its own off-grid median, so that dip depth is comparable across geometries and signal families.

Three signal modes are swept. On a pure sinusoid the collapse is exact and z reaches zero at a locked frequency; on a Light TSMixup or KernelSynth background the background breaks patch identity, so the collapse appears as a deep local minimum instead. Fitting all three is what shows that the conclusion is not an artefact of noise-free inputs.

The fit is a regression of the log profile on two indicator labels,

$$\log z_g(f) \sim \mathcal{N}(\alpha_g + \theta_S \cdot \mathrm{OnStrideGrid} + \theta_P \cdot \mathrm{OnPatchGrid},\ \sigma)\,, \tag{48}$$

where a frequency counts as on a grid when its distance to the nearest member of that comb is within tolerance. The logarithm makes the labels multiplicative on the per-geometry off-grid level α_g, so θ_S and θ_P are depths relative to that level rather than absolute values, and geometries whose dispersion sits at different scales remain comparable. Four labellings are fitted to the same profiles and compared by LOO — stride only, patch only, both, neither — and the comparison is meaningful only because the sweep grid is the union over every geometry, so each labelling is scored where its rival predicts a dip as well as where it does. It is identified only because the design contains configurations with $S \nmid P$: on the $P = S$ diagonal the two combs coincide by construction and no amount of data can separate them.

Model D2, the movement of the sites

Where D1 says where the dips sit, D2 asks whether they move. Sites are detected on each swept profile, adjacent detections are merged so that one dip straddling two grid points is not counted twice, and each site is assigned to the branch that predicts it; sites predicted by both branches are set aside, because $\mathcal{F}_{\mathrm{lock}}$ is a union and the union of two combs has two interlaced spacings rather than one. The fundamental of each branch, the lowest of its detected sites, is then regressed on the spacing that branch predicts:

$$\hat{f}^F_{1,g} \sim \mathcal{N}\left(\kappa_F\, \Delta^F_g,\ \sqrt{\sigma_F^2 + \Delta f^2}\right), \qquad \Delta^S_g = \frac{f_s}{S_g}, \quad \Delta^P_g = \frac{f_s}{P_g}. \tag{49}$$

The slope κ_F is the estimand: measured spacing per unit of predicted spacing, so $\kappa_F = 1$ is a branch that tracks its own arithmetic one for one, and its prior is centred on 0, no relationship, so the hypothesis has to be earned from the data. The residual carries the floor Δf, the sweep step, because a spacing read off a discrete grid is no more precise than one step; without it, sites landing exactly on the prediction drive σ_F to zero and the posterior develops a funnel the sampler cannot traverse. With the floor, a near-zero σ_F is a result rather than a sampler failure.

Both the fundamental and the median gap between successive sites are fitted. With three or four sites in band a single spurious detection inserts a short interval and flips the median, to which the fundamental is immune, so a disagreement between the two is itself informative about the stability of the detection.

Appendix E: The Exploratory Frequency Sweep

The two figures below are the descriptive pass that precedes the Bayesian analysis, run on pure sinusoids swept from 2 to 250 Hz in one-hertz steps at $f_s = 512$ Hz over the five geometries of the initial design. They are reported here rather than in the body because neither is a hypothesis test: both average over phase, use a single signal family, and carry no statement of uncertainty. Their role is to show that the phenomenon the models of Bayesian Analysis are built to measure is present in the raw sweep. The summary statistics read off them are given in Bayesian Results.

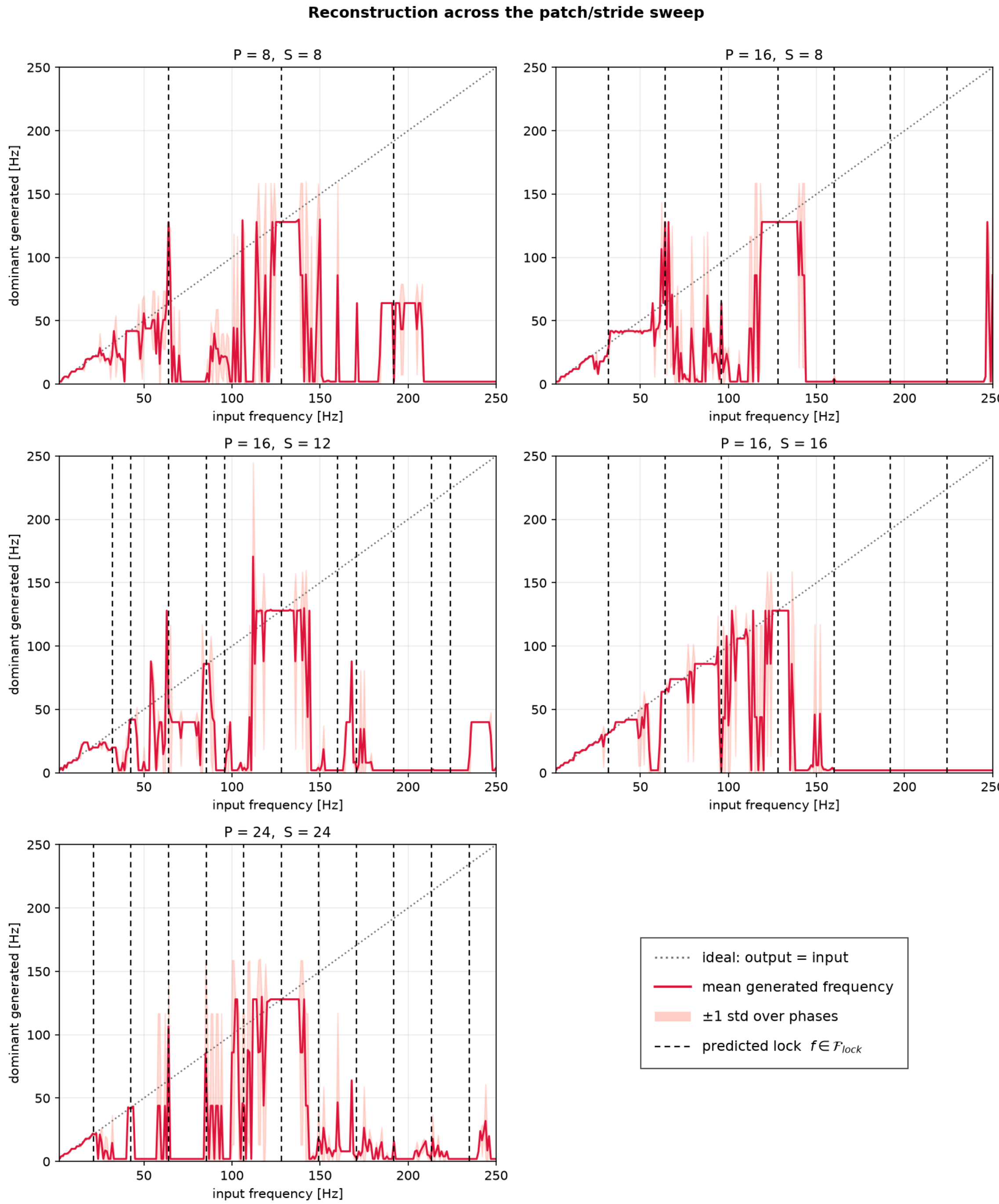

Figure 6: *Behavioural reading. Each panel plots, for one geometry, the dominant frequency of the recursive multi-step forecast against the frequency of the tone supplied as context, averaged over the non-redundant starting phases with one standard deviation shaded. The tone is given as context, the model's median forecast is appended to that context and the model re-invoked, and the loop continues until a full window has been generated; the dominant frequency is then read as the largest magnitude of the discrete Fourier transform after the mean is removed. The recursion is imposed by this procedure and is not the model's own generation mode, since Chronos-Bolt emits its whole horizon in one step. A point on the dotted diagonal is a tone rebuilt correctly; a flat segment is a set of distinct inputs collapsing onto one output; a jump is a tone the model rebuilds that was never supplied. Dashed verticals are the predicted candidates* $\mathcal{F}_{\text{lock}}$ *of that geometry. The reading is behavioural and records what the model outputs, not what its representation retains.*

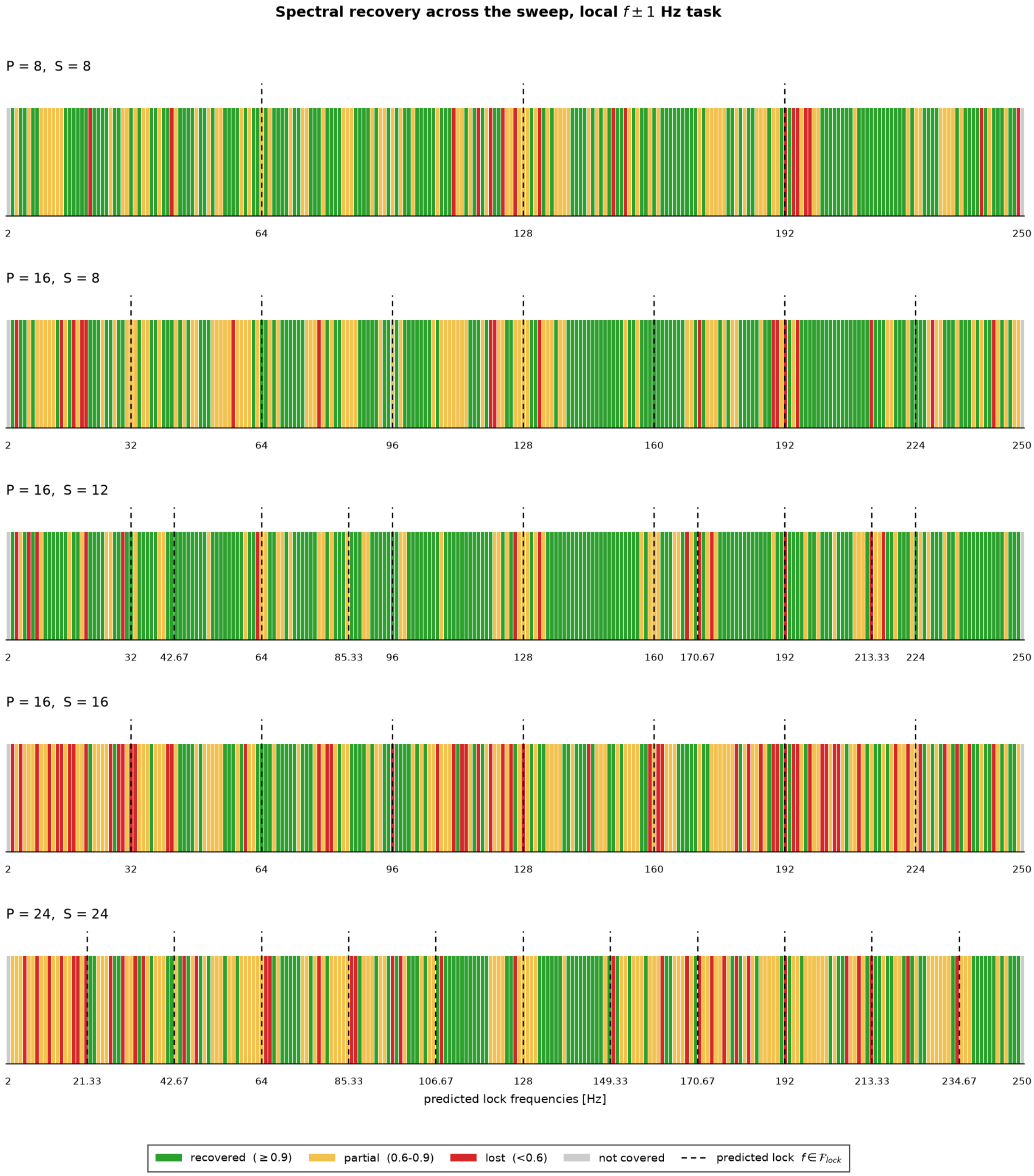

Figure 7: *Representational reading. Each strip is one geometry, and each cell one swept frequency, coloured by the cross-validated accuracy of a linear probe trained on the internal state at the output quantile head. The task is local and of constant difficulty at every frequency, whether a tone at $f - 1$ Hz can be separated from one at $f + 1$ Hz, and it holds out phases rather than frequencies, so no cell is inflated by a probe memorising individual frequencies. Green is accuracy at or above* 0.9, *amber between* 0.6 *and* 0.9, *red below* 0.6, *which is close to chance on a two-class task; grey marks a frequency the task does not cover, which is a statement about the design and not about the model. Dashed verticals are the predicted candidates of that geometry. Read against Figure 6, a frequency that is off the diagonal but still green has lost the tone at the projection rather than at the tokenisation; a red cell is the stronger condition that structural aliasing predicts.*